

Being A Researcher

Comprehensive Guidelines for Early Researchers

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Research Area: Self-powered Sensors, Energy Harvesting,
Nanogenerators-based Biomedical Devices

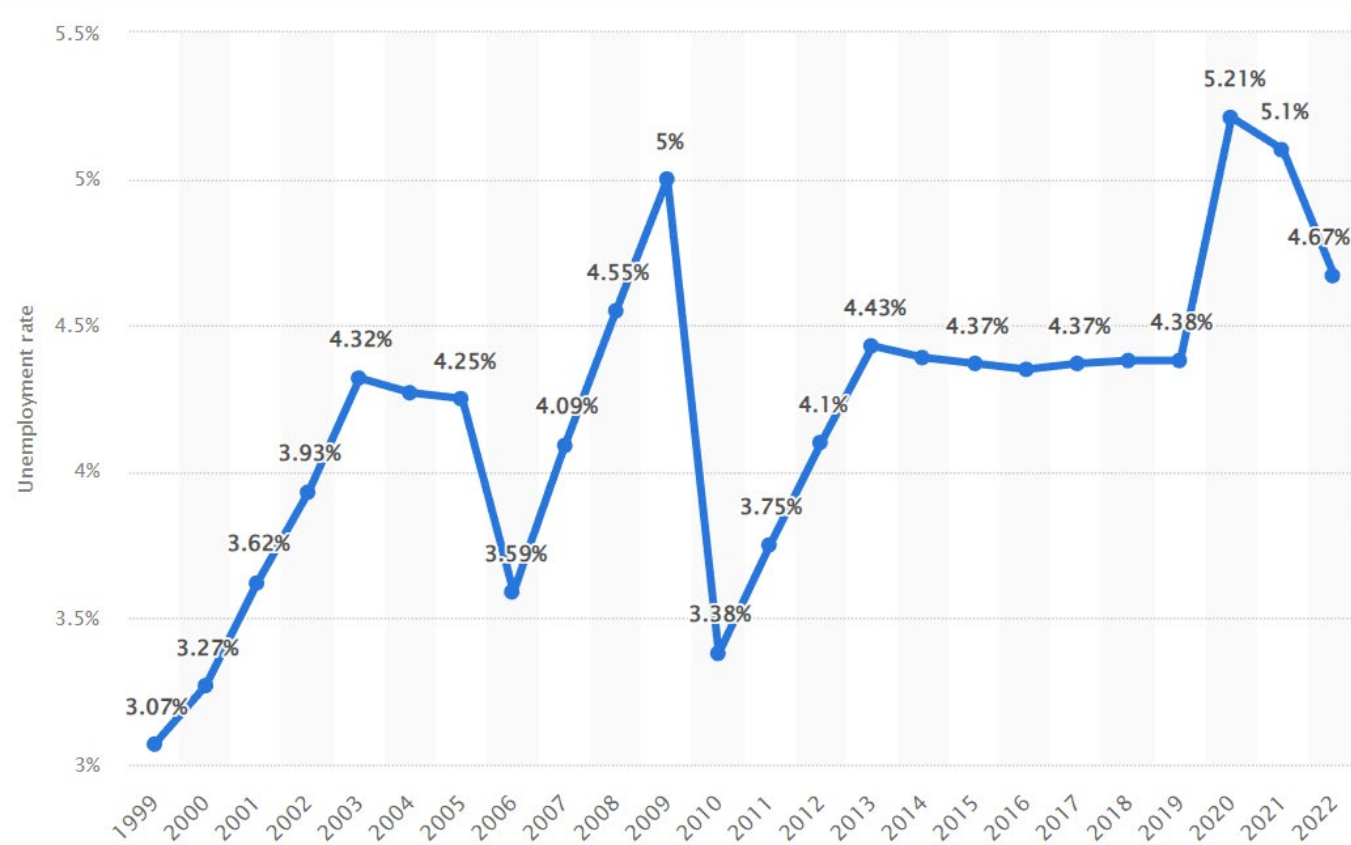
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Content

- ❑ Introduction
- ❑ Research experience in higher studies
- ❑ Why should we move toward Ph.D.?
- ❑ Diving into the research work
- ❑ Summary of the talk

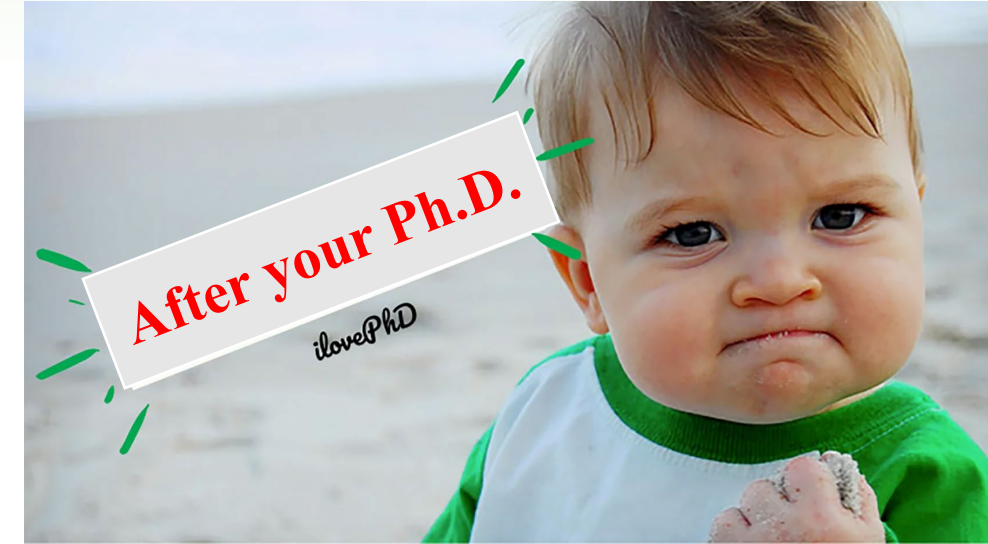
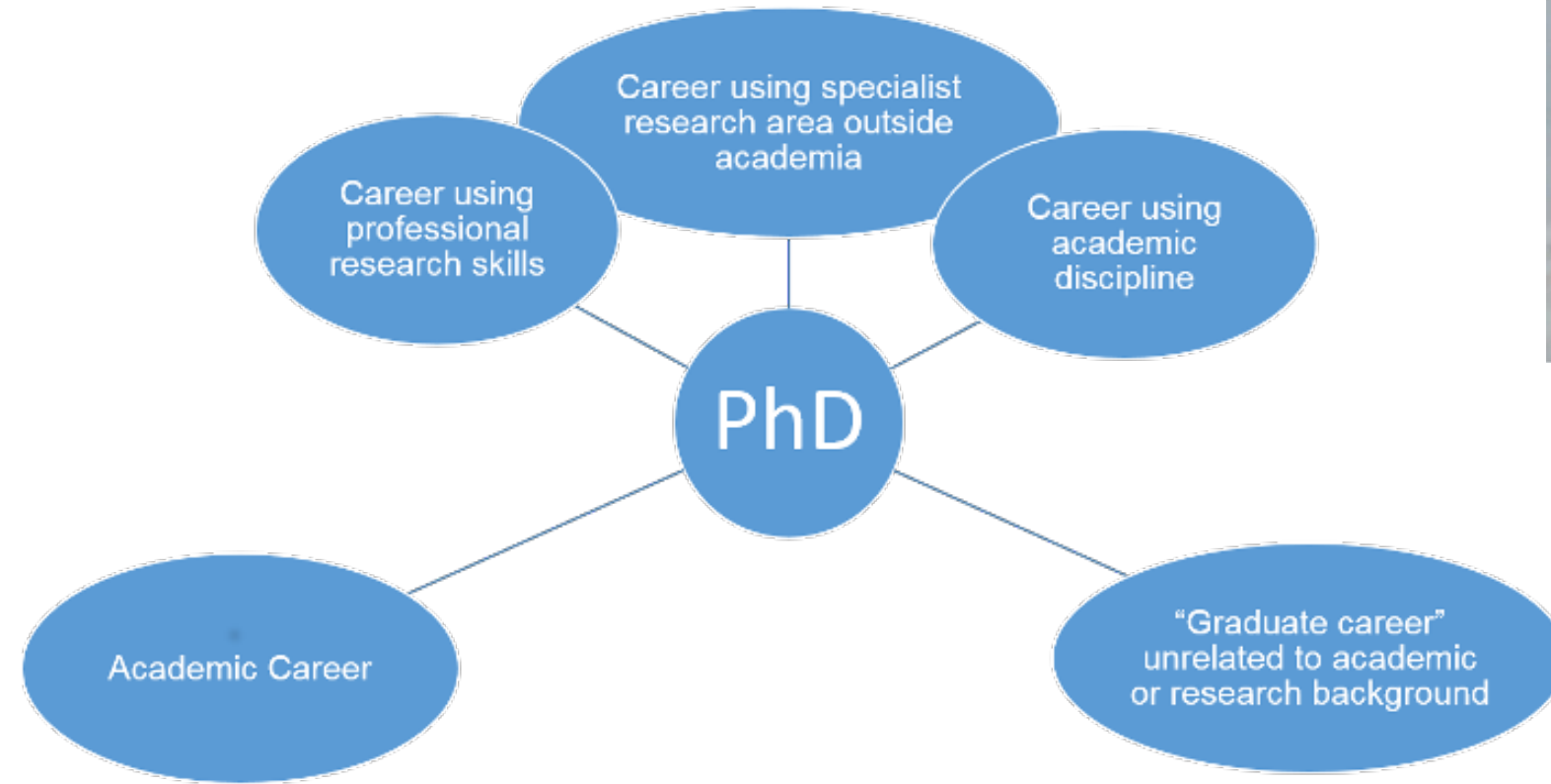
Perspective: Bangladesh

- Domestic job crisis
- Limited resources
- Unwillingness in research
- Adverse situation
- Faulty education system
- Dirty manipulations
- Age-old traditional concept
- Immoral working culture
- Lower salary



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Perspective: After your Higher Study



- No job limitation
- International Career
- Self-improvement
- Diverse Recognition
- Skilled worker
- Handsome salary
- Quality life

Nothing is easy!

According to Nature's 2016 jobs survey,

❑ An average academic researcher spends fewer than four out of every 10 working hours on research.

❑ Time in the lab shrinks as you progress your career, with researchers aged 45 years or older.



38%
Research



16% Writing
papers



10% Writing
grant applications



11% General
administration

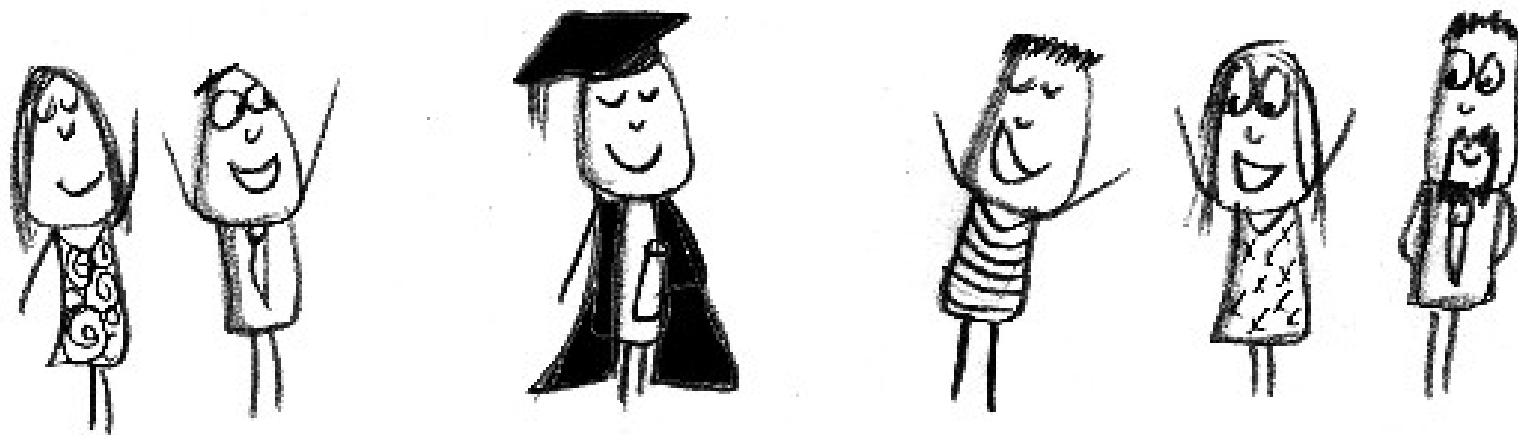


5% Data
storage



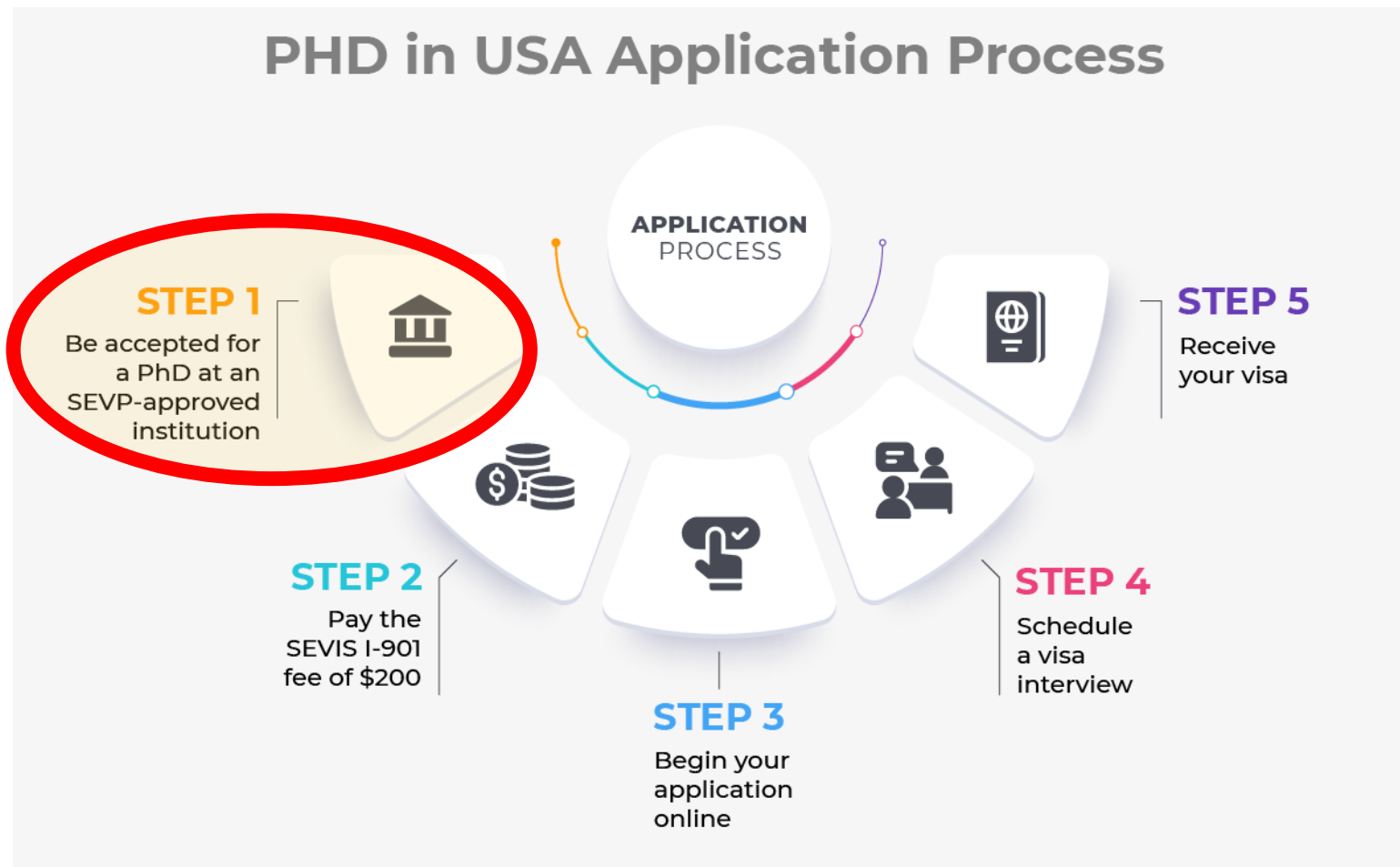
20% Teaching

Research experience in higher studies



Stepping into the tract

- Academic performance
 - Research experience
 - Research proposal
 - GRE/IELTS
 - Personal statement
- Letters of recommendation
 - Funding organization
- Extracurricular activities



What is Research experience ?



Research experience Includes

Lifetime Experiences

- Establishing a research goal
- Designing and conducting experiments
- Analyzing obtained data
- Writing research articles
- Presenting findings at conferences
- Publications in journals
- Collaborating with other researchers
- Securing funds for research projects

Experiences in daily works

- ✓ Literature reviews
- ✓ Developing hypotheses
- ✓ Collecting and interpreting data
- ✓ Using research tools
- ✓ Implementing specific methods

Evolution of research experience

	Elicit ideas	Add ideas	Distinguish ideas	Reflect
Develop practices	Identify or formulate a question in the context of the lab's research goals	Conduct experiments, collect and organize data	Analyze and interpret data Evaluate evidence Critique conclusions	Make final conclusions and plan next steps
Expand content knowledge	Articulate hypotheses and questions about the research topic	Read literature, attend seminars, discuss with research team	Consider quality of evidence and relevance to argument	Synthesize experimental results
Understand nature of science	Express expectations for science research experience	Attend lab meetings, experience experimental failure	Present progress reports and compare ideas in group setting	Consider how discoveries emerge from iterative processes
Develop identity in science	Share goals for the URE relative to personal and career aspirations	Participate in social network of research team	Experience how process of criticism contributes to research progress; share ideas as a team	Recognize strengths related to career aspirations

Research for extending other expertises

Research experience such as working on research projects, publishing papers, or presenting findings at conferences, can demonstrate your ability to conduct independent research and make meaningful contributions to your field.

A personal statement by crafting a compelling personal statement that highlights your academic background, research interests, and future goals can help you stand out from other applicants.



Research for extending other expertises

Obtaining strong letters of recommendation from respected professors, researchers, or professionals in your field can help demonstrate your academic potential, research capabilities, and suitability for graduate-level study.

Ensuring a good fit between your research interests and the priorities of the funding organization can increase the chances of receiving funding.

Developing a clear and well-articulated research proposal that aligns with your research interests and fits within the funding organization's priorities is essential.



Benefits from Research

ACADEMIC

Contribution to advances across and within disciplines, including significant advances in understanding, method, theory and application.



CULTURAL

Contribution to people's understanding of ideas and reality, values and beliefs.



ECONOMIC

Contribution to a company's costs and revenues (micro level), and economic returns through increased productivity or economic growth (macro level).



EDUCATIONAL

Contribution to education, training and capacity-building, including through curricula, educational tools, and qualifications.



ENVIRONMENTAL

Contribution to managing the environment, such as protecting natural resources, reducing environmental pollution, improving weather forecasting, and tackling the climate crisis.



HEALTH

Contribution to public health, life expectancy, health-related quality of life, prevention of illness, and reduced health inequality.



POLITICAL

Contribution to how policymakers act, to how policies are constructed, and to political stability.



SOCIAL

Contribution to community welfare and quality of life, and to behaviours, practices, and activities of people and groups.

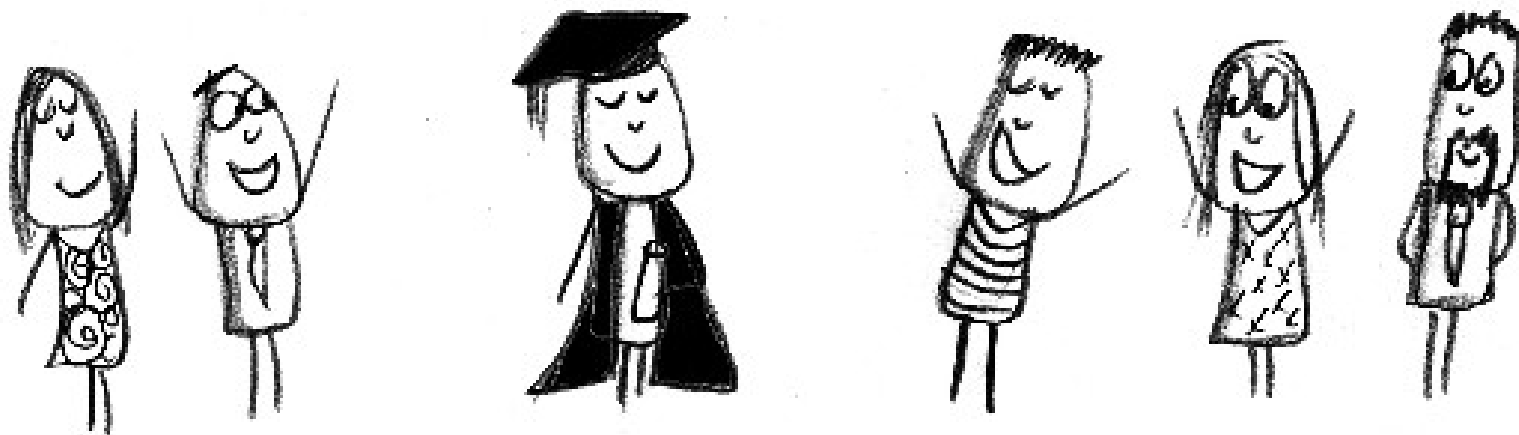


TECHNOLOGICAL

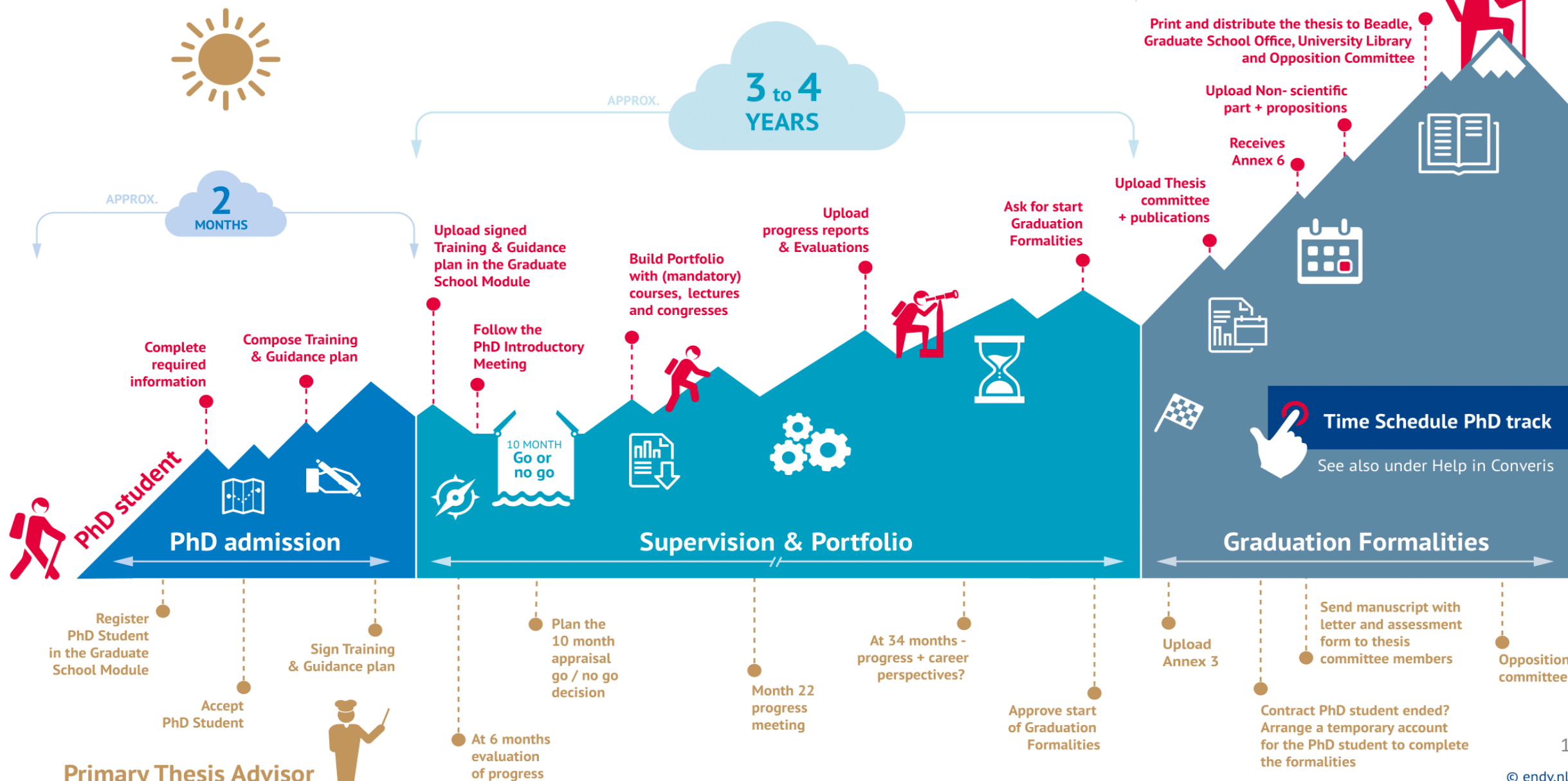
Contribution to the creation or improvement of products, processes and services.



Why should we move toward Ph.D.?



*Pursuing a Ph.D. requires a substantial commitment of **time, energy, and resources**. However, the potential **personal and professional benefits** make it a worthwhile endeavor.*



Creating Knowledge

- ❑ One of the primary reasons is to contribute to the advancement of knowledge in their field of study.
- ❑ By conducting original research and publishing their findings, Ph.D. graduates have the opportunity to expand the boundaries of what is known and discover new insights into complex problems.
- ❑ Pursuing a Ph.D. provides an opportunity to develop expertise and contribute new knowledge to the field.



KNOWLEDGE

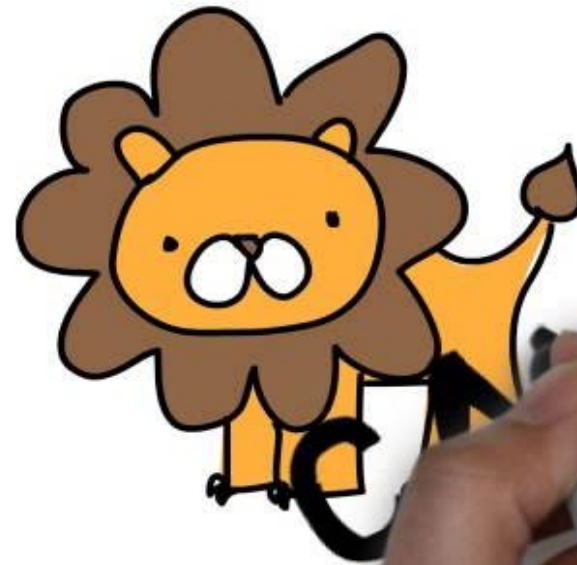
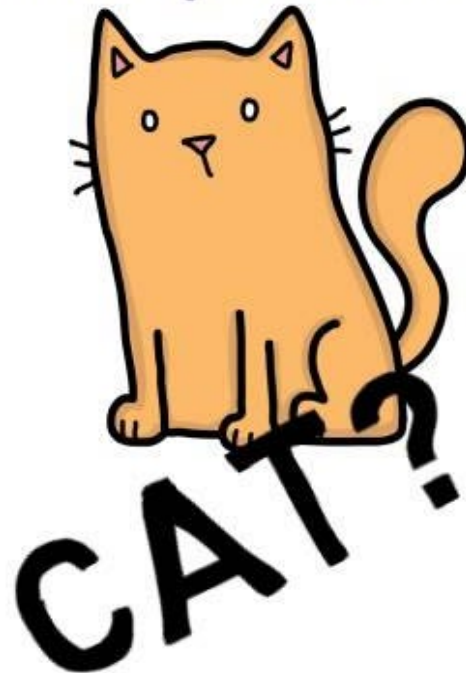
Triggering Intellectual stimulation



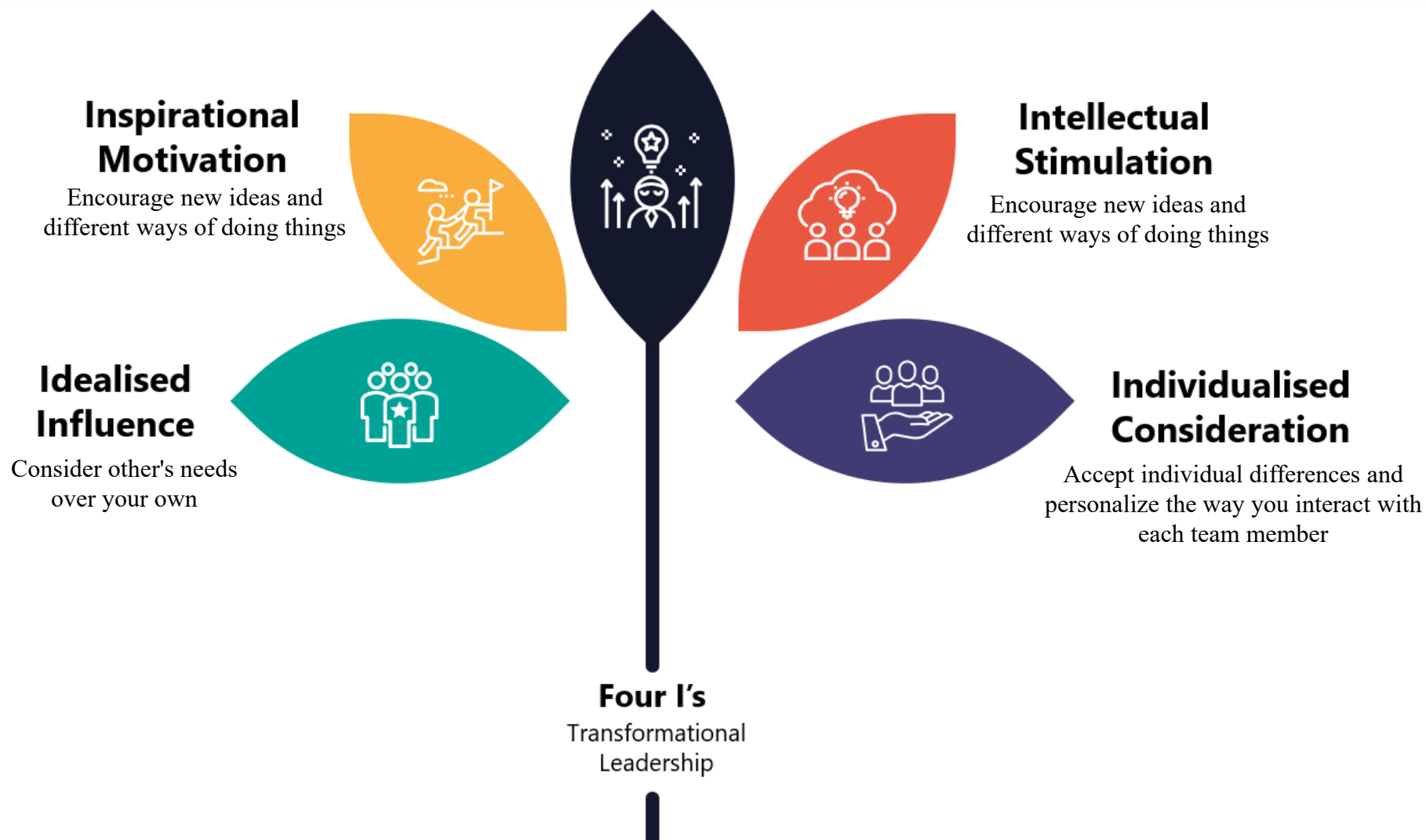
Handling Intellectual Challenges



The challenge can come because there are no clear distinctions between things or different interpretations are possible.

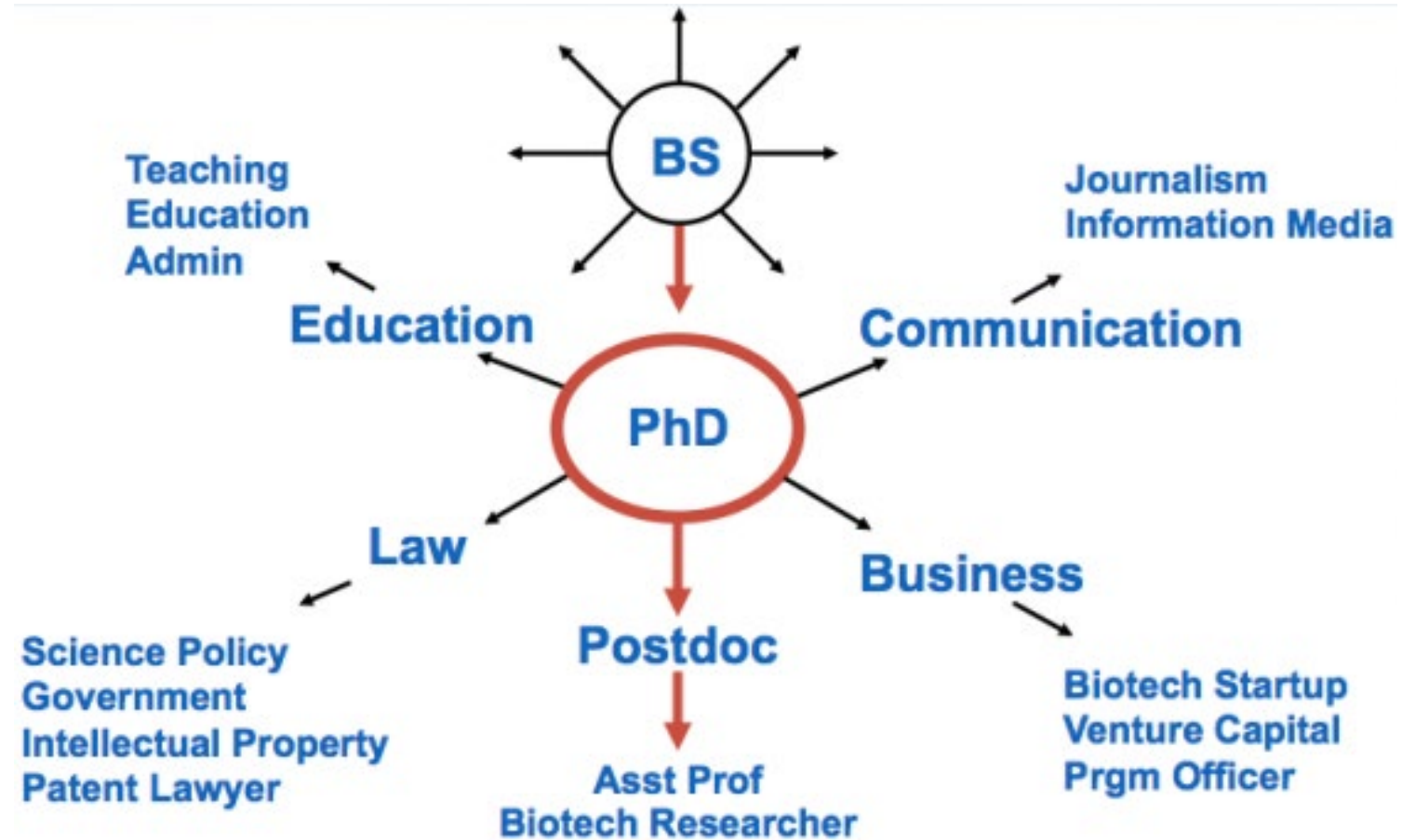


Building up Leadership



Career Advancement

- ✓ A Ph.D. can open up opportunities for career advancement in **academia**, **research organizations**, **government agencies**, and **industry**.
- ✓ Many positions require a Ph.D., and having one can increase your chances of obtaining **higher-paying**, **leadership roles**.



Personality development



THE PERSONAL GROWTH MODEL

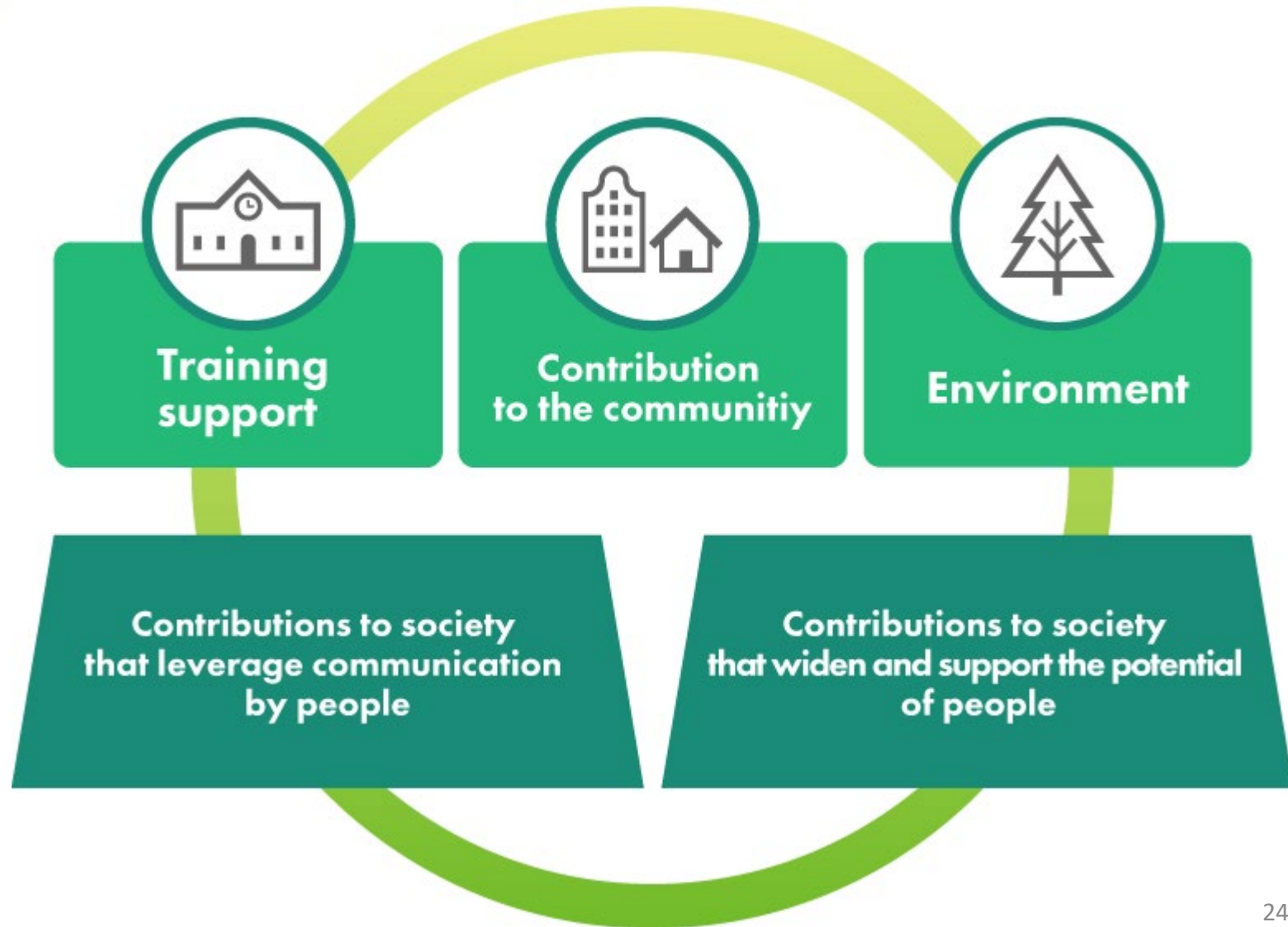


A Ph.D. can be fulfilling and satisfying

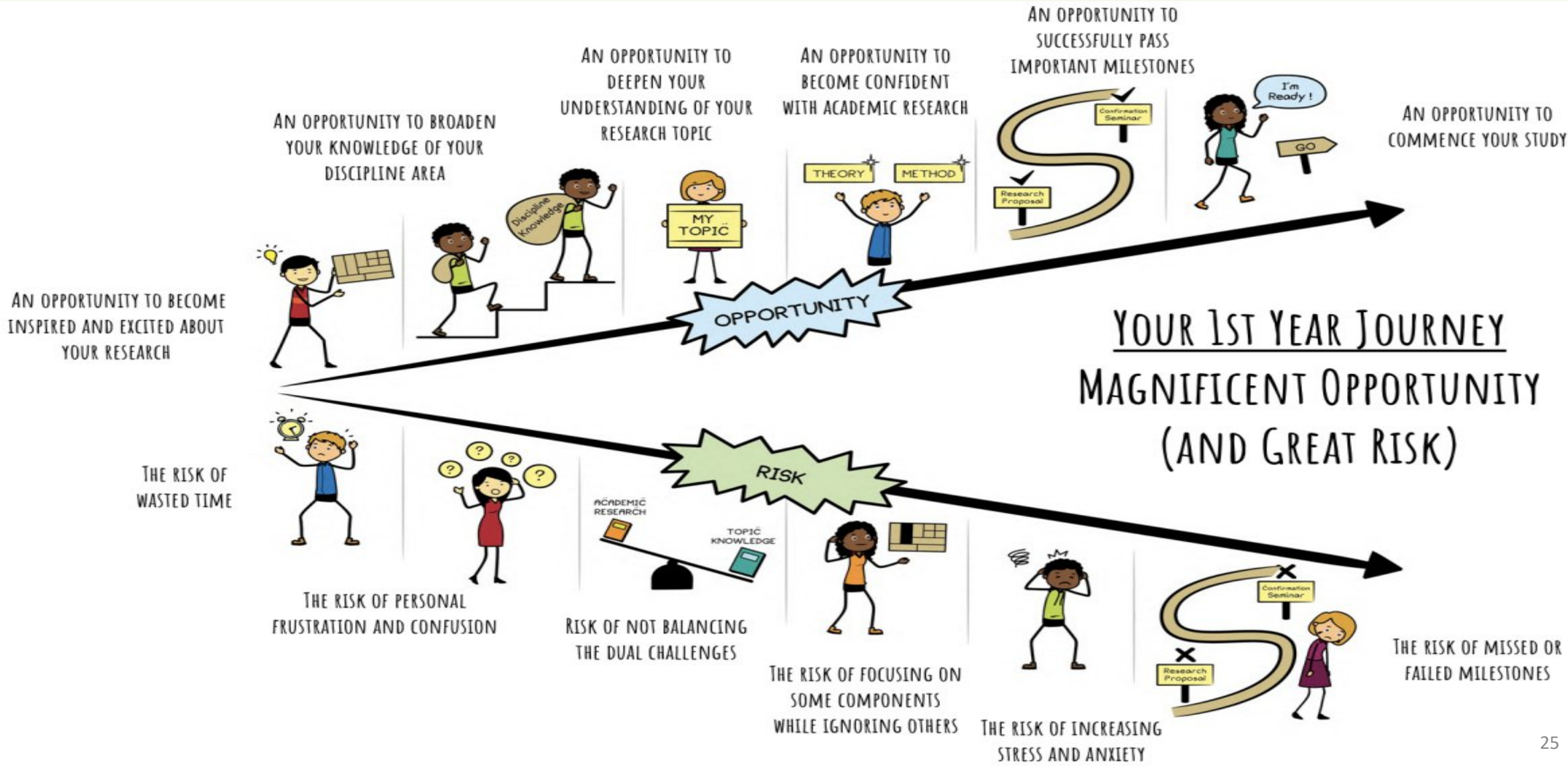


Contribution to the society

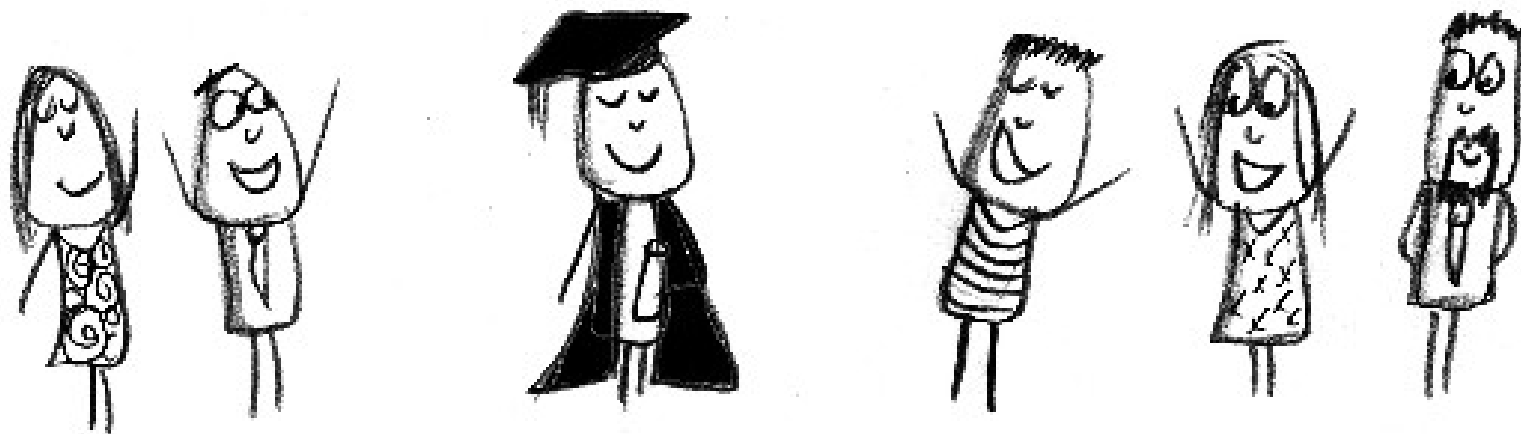
The research conducted during a Ph.D. program has the potential to contribute to society by **providing new insights, solutions, and innovations** to **address pressing issues and challenges** in the respective community and society.



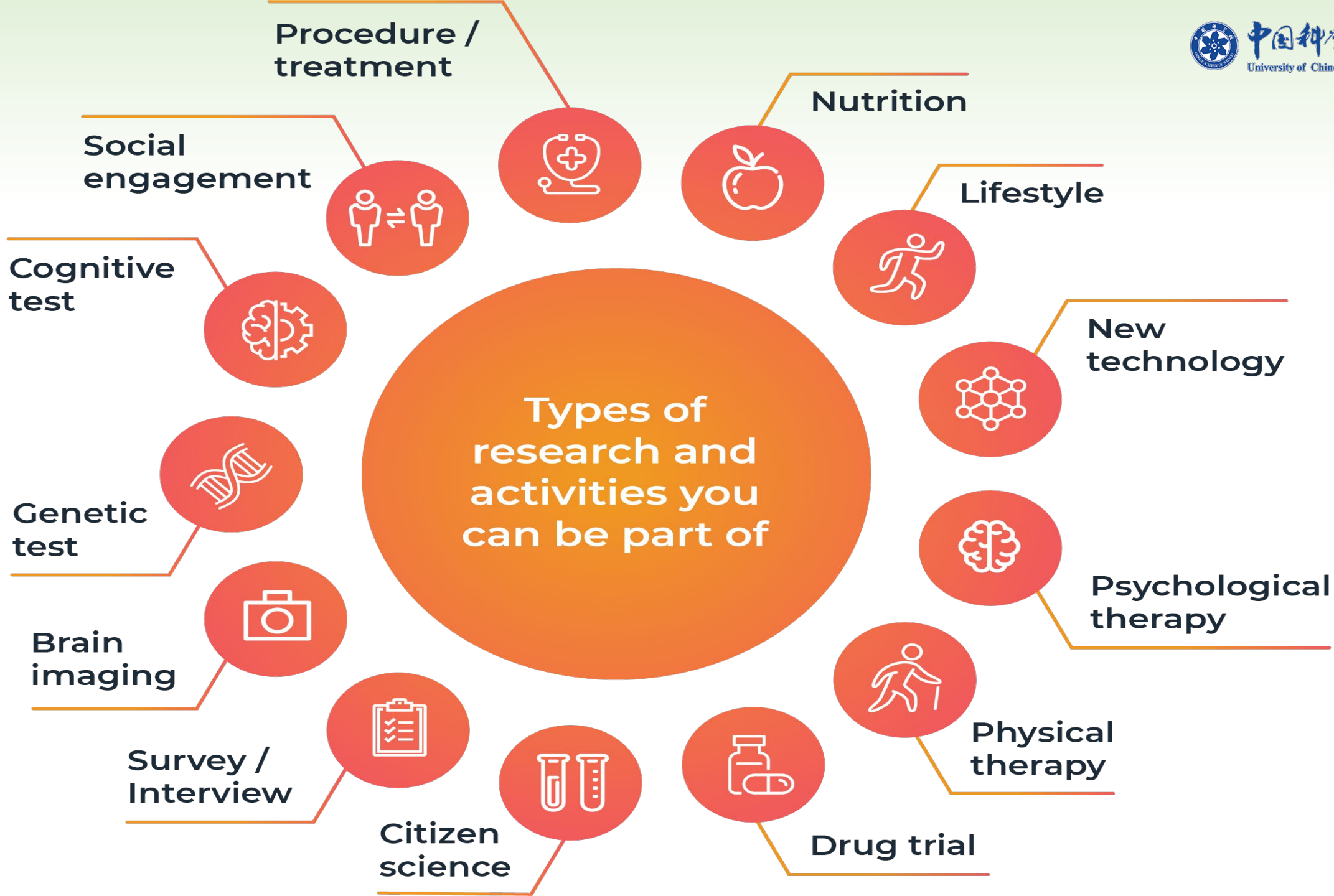
Underline Risks



Diving into the research work







Choosing a good topic

Select a topic that interests you and aligns with your area of study. For example, if you're studying biology, you might choose a topic related to genetics, ecology, or molecular biology.

- ✓ It interests you
- ✓ There is the availability of materials to support it
- ✓ It is not over-exploited
- ✓ You can manage to work on it using available resources
- ✓ It meets the professor's requirements



Characteristics of a good topic

- The topic should be **rigorous** – to maintain quality
- It should be **verifiable or provable** using data – qualitative or quantitative data or both
- Research topics should be **controllable or measurable** – like weight and age
- It should be **clear, concise, and valid** – avoid vagueness
- A topic should also be **original, sequential, and coherent**
- Lastly, your research topic should be **accurate**



Refining your topic

- **HOW** did you choose the topic? What interests you about it? Do you have an opinion about the issues involved?
- **WHO** are the information providers on this topic? Who might publish information about it? Who is affected by the topic? Do you know of organizations or institutions affiliated with the topic?
- **WHAT** are the major questions for this topic? Is there a debate about the topic? Is there a range of issues and viewpoints to consider?
- **WHERE** is your topic important: at the local, national, or international level? Are there specific places affected by the topic?
- **WHY** is/was your topic important? Is it a current event or a historical issue? Do you want to compare your topic by time period?



Importance of Literature Review

Conduct a thorough review of relevant literature on your topic. This involves finding and reading academic articles, books, and other sources of information related to your research question or issue.



Literature Review Process



Preparation

- What is your topic?
- What resources do you have?
- How will you keep track of your research?



Retrieval

- Brainstorm search terms
- Use filters to narrow
- Keep a record of searches you've tried



Appraisal

- Skim the articles you're finding
- Do they look relevant?
- Do they meet your criteria?



Synthesis

- Read and take notes
- Notice any themes emerging across sources
- Pay attention to what other articles are cited



Write-up

- Consider audience for format, length, tone
- Can group articles by theme
- Include analysis/critique



Sharing

- How to reach your target/imagined audience?
- Are there others who might be interested?
- What are the best avenues to share?



Future

- Document your work along the way so your process can be replicated
- Make sure your work is safely stored so it can be accessed later



This process doesn't have to be linear!
You can return to each step as needed throughout your research.

Potential objective goals

Define your research question

- Inclusions
- Exclusions

Identify relevant search terms

- Scoping searches
- Terms in related reviews

Develop your search plan

- Combine terms with AND / OR, or as phrases
- Apply truncation / wildcards

Test and refine your search

- Check results for key articles
- Apply limits or filters

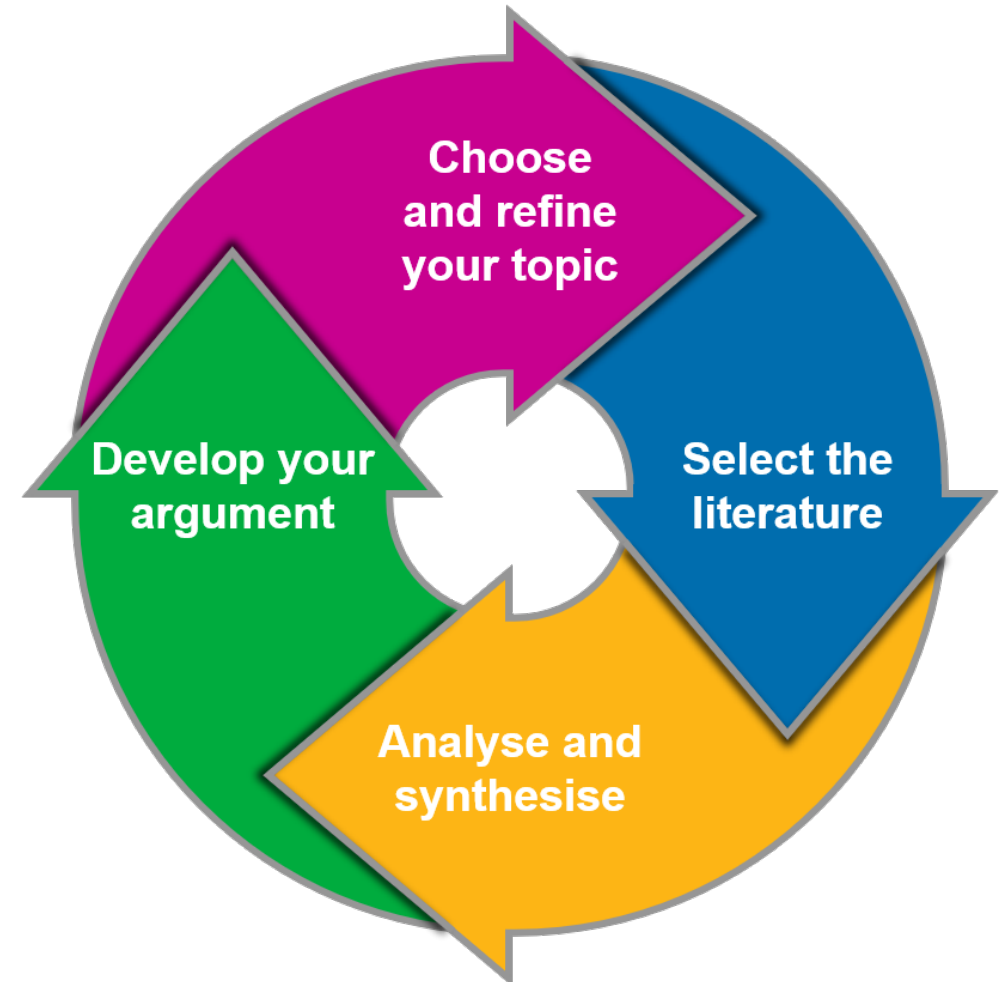
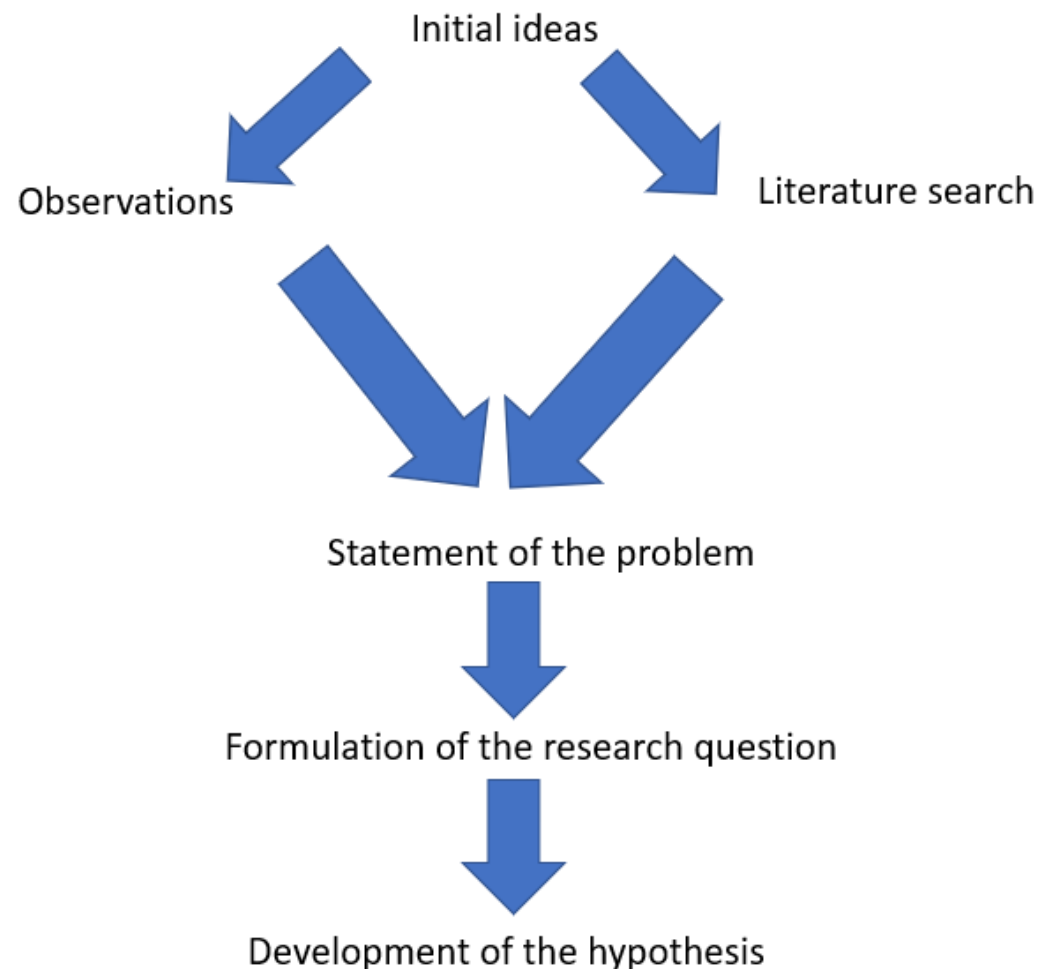
Select relevant databases and Run searches

Document and manage search results

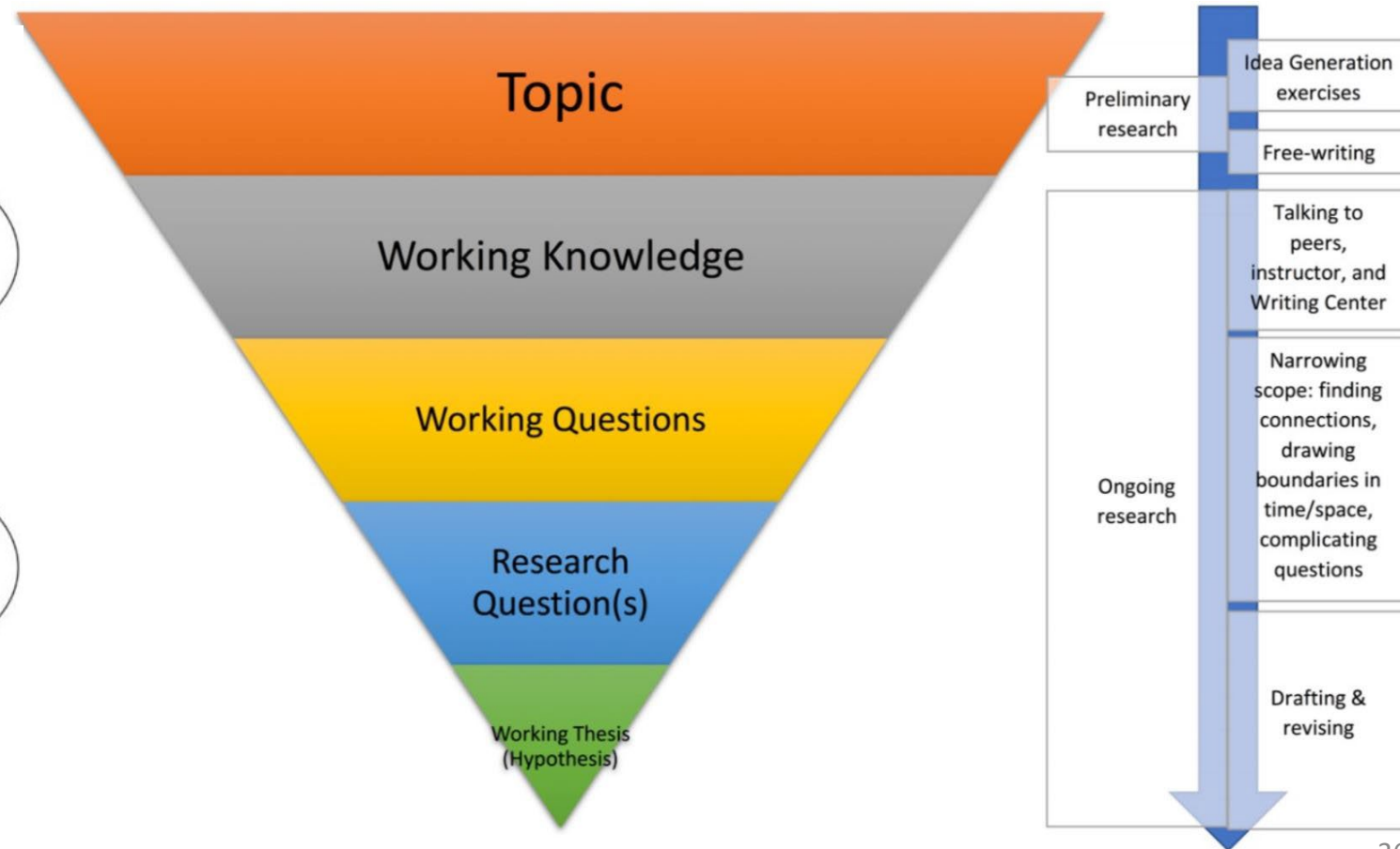
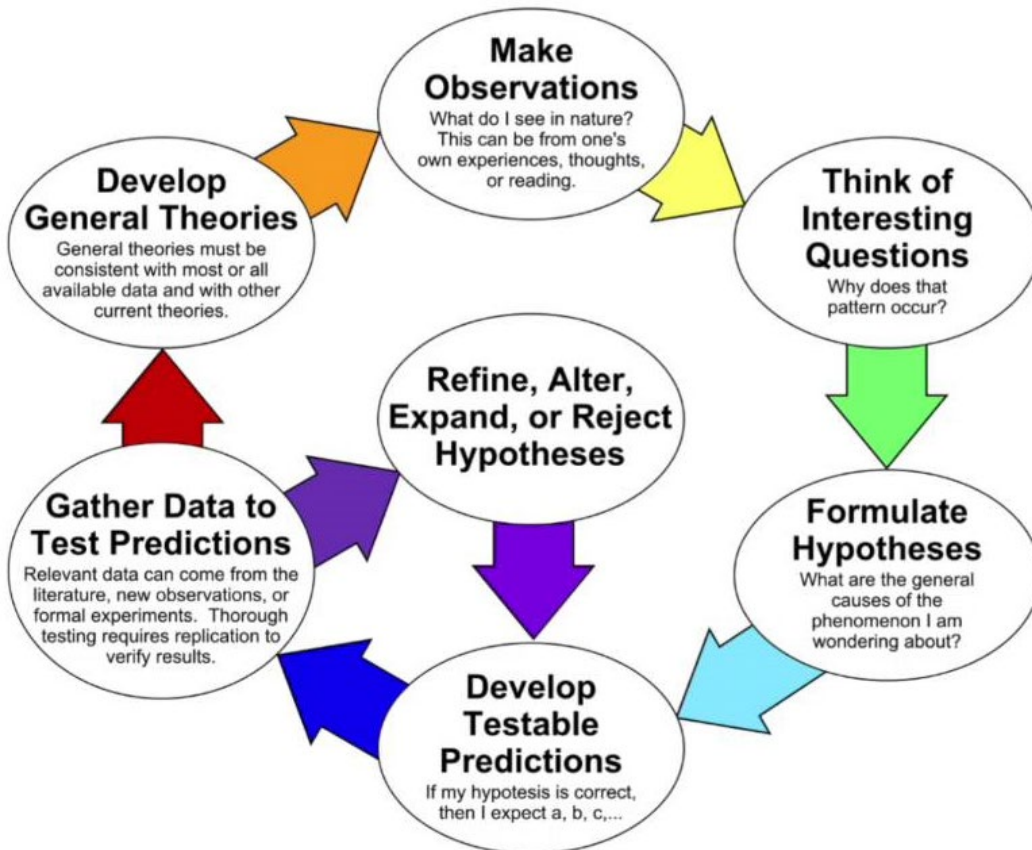
- Save searches
- EndNote

Developing research questions or hypotheses

A research question is a broad inquiry that you seek to answer through your research. In contrast, a hypothesis is a specific statement of prediction that you test through your research.

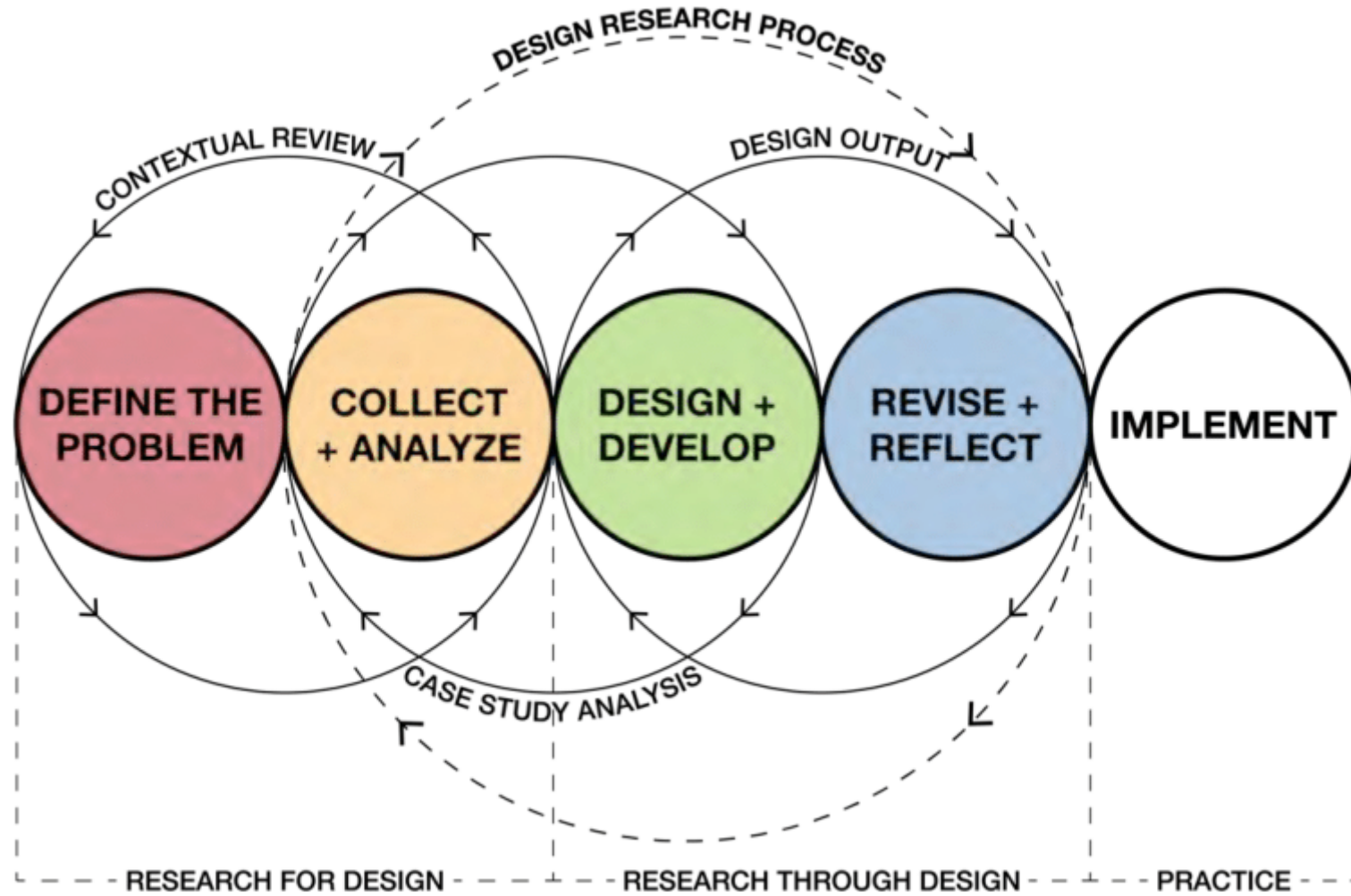


Studying the hypotheses

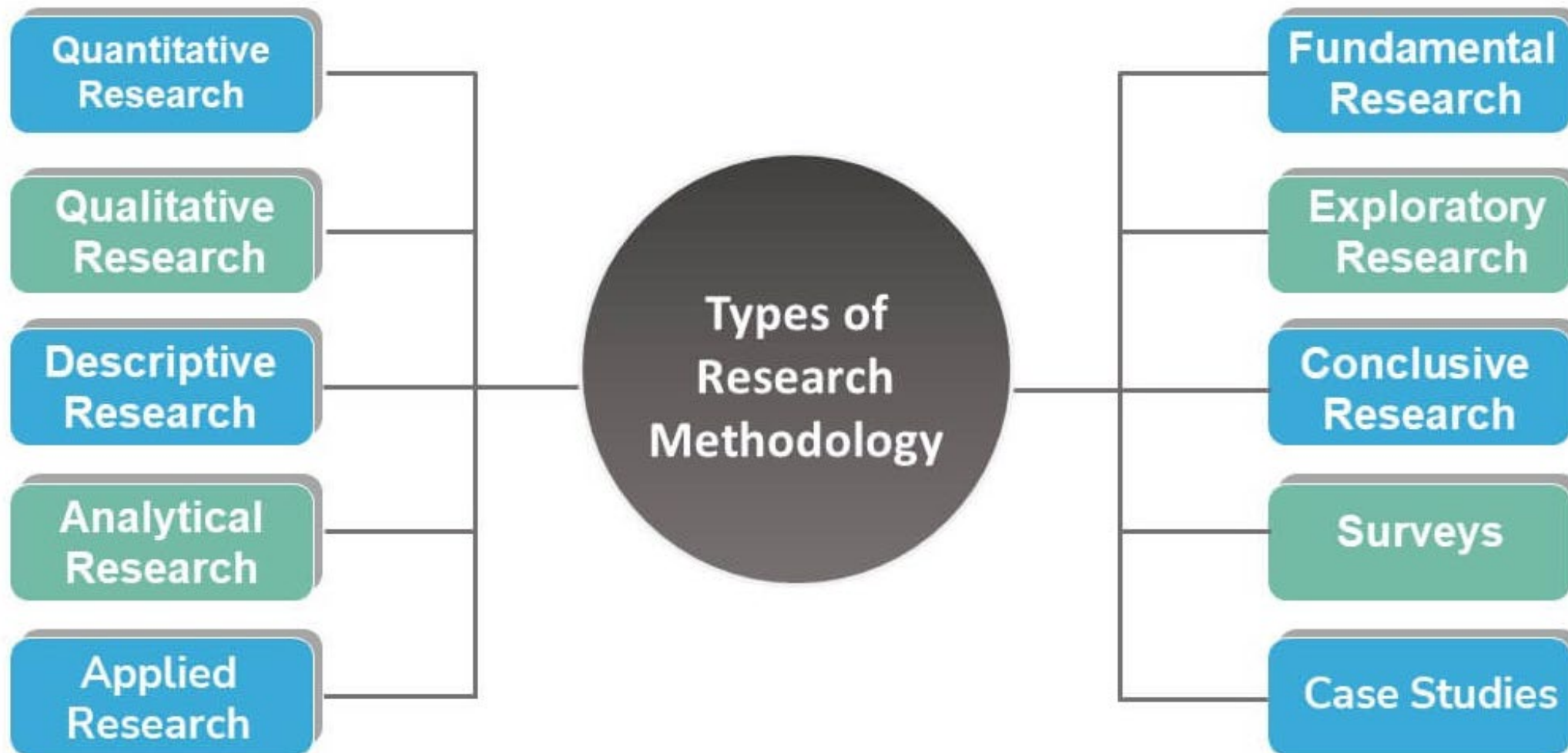


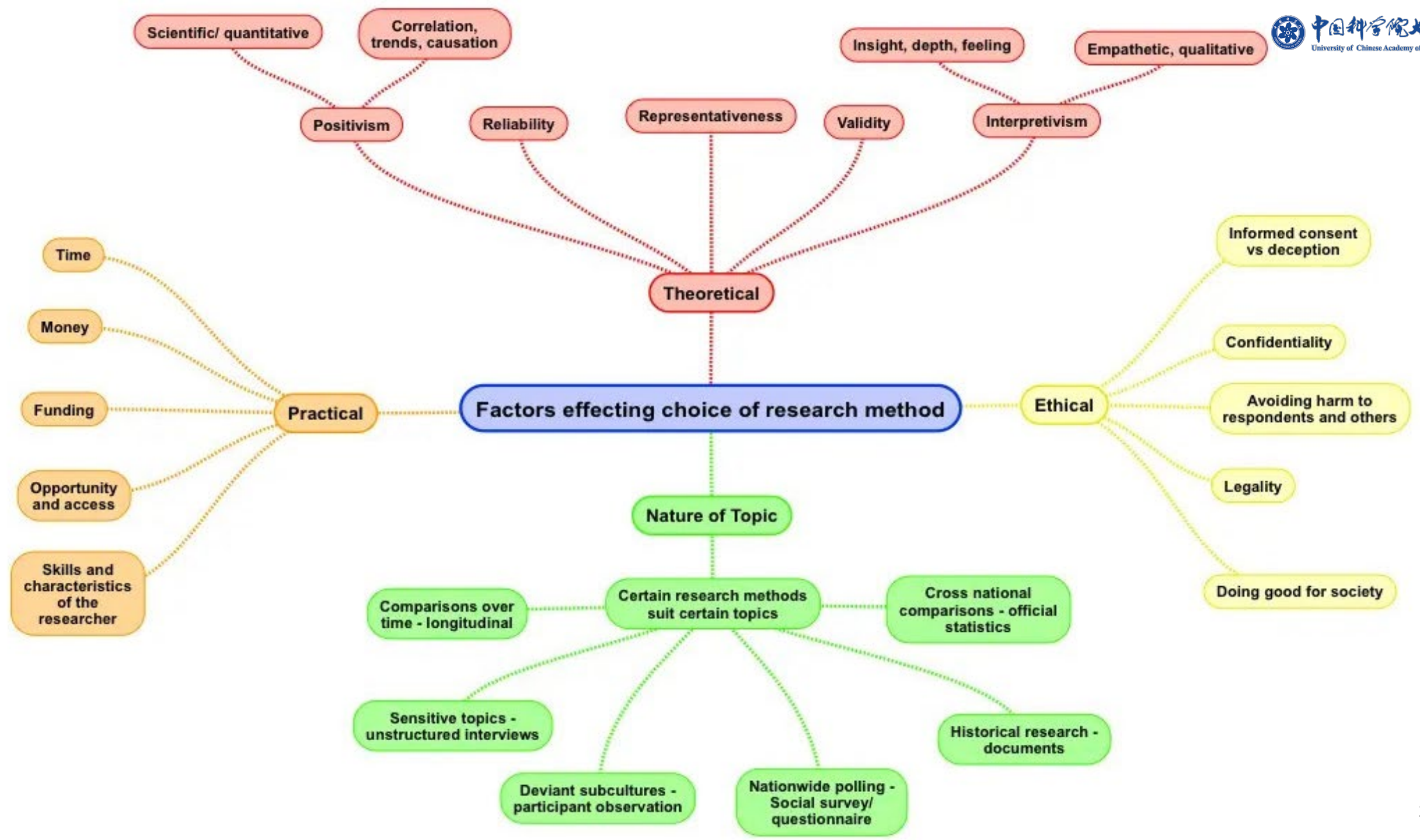
Design Research Methodology

Select a research methodology that aligns with your research question or hypothesis. This could involve conducting experiments, surveys, case studies, or ethnographic research.



Types Research Methodology



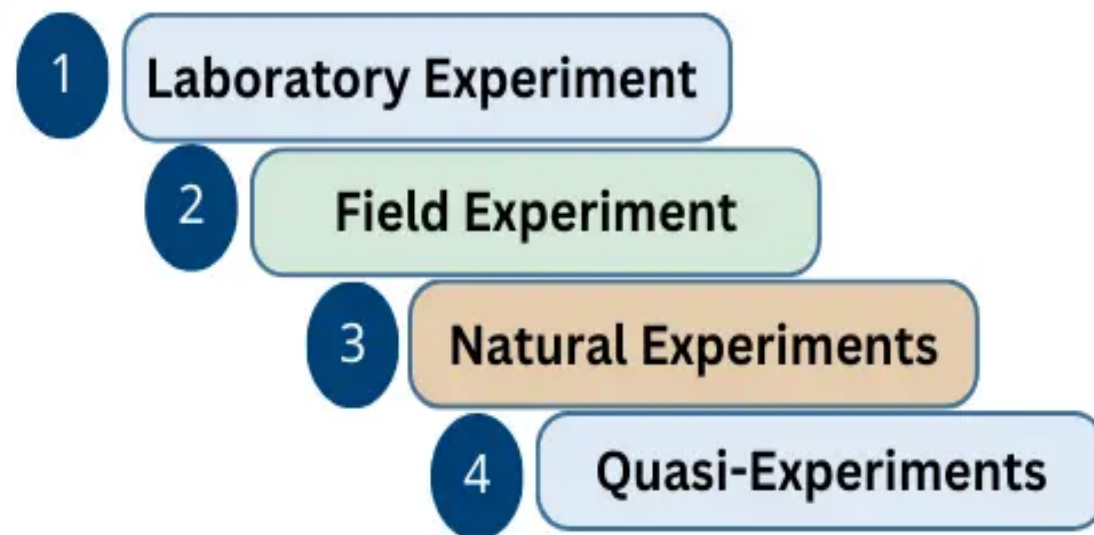


Data collection

Collecting data through the methodology you have chosen is the most important step in the research as the result of the results completely depends on the proper data collection. To get data in STEM fields it is important to work in the labs with proper devices and if you've decided to conduct a survey, you'll need to develop a questionnaire, recruit participants, and collect their responses for non-STEM subject areas.



Non-STEM subjects



STEM subjects

Points to be noted

- Be sure to learn the basic procedures for using scientific instruments.
- Follow the proper etiquette during the surveys.
- Grant required permission for collecting sensitive data from the authority.
- Maintaining data security is very important.
- Data manipulation must be avoided to fabricate result data.



Analyzing Data

Once you have collected your data, you will analyze it using appropriate statistical or qualitative analysis methods that align with your research methodology.



Descriptive Analysis



Inferential Analysis



Predictive Analysis



1. Define your questions carefully



2. Establish measurement priorities



3. Collect all the relevant data.



Prescriptive Analysis



Text Analysis



4. Analyze the data you gathered



5. Review and interpret the results.

Data analyzing and presenting tools

- ❑ Write, review, and publish documents tools: Word, Docs, yWriter, Lyx, Overleaf, Texmaker
- ❑ Improve grammar, spelling, or writing style tools: Grammarly, Prowriting aid, Hemingway, Antidote, I checker, Slick write
- ❑ Reference management tools: Mendeley, End note, Citavi, Zotero, Jab ref
- ❑ Presentation and Poster Design: PowerPoint, Publisher, Indesign, Illustrator, Photoshop, After Effects, Blender Narton, Canva
- ❑ Data Analysis and Graphing: Origin Pro, Magic Plot, SciDAVIs, LabPlot, GunPlot
- ❑ Advanced Statical Analysis: SPSS, Statgraphics, Prism, R, Jamovi, PSPP
- ❑ Chemical Structure Drawing Software: ChemDraw, ChemDoodle, Chems sketch, Marvin, Bkchem, JChemPaint



Write Research Article

Finally, write up your research paper, including an introduction, literature review, methodology, results, discussion, and conclusion sections. Make sure to follow appropriate formatting and citation guidelines.

- ✓ Understand your goal
- ✓ Choose your topic
- ✓ Research your topic
- ✓ Build an outline and thesis statement
- ✓ Write your first draft
- ✓ Cite your sources
- ✓ Edit and proofread



Guess What? - Introduction

- What are you trying to say, prove, argue, establish
- This is stated in the introduction.
- This is also stated in each topic sentence of the body of the essay.



Prove it! - Body

- How can I back it up?
- What evidence will I use to substantiate my point of view.



So What? - Conclusion

- In the conclusion, relate your findings to your main argument.
- In the body of your essay at the end of each paragraph, relate your findings to your main argument.

Manuscript of the Research Article



How to Structure
Your Manuscript?



- 
- 1 Impactful & descriptive **Title**
 - 2 Structured/unstructured **Abstract**
 - 3 Discoverable **Keywords**
 - 4 Explanatory & powerful **Introduction**
 - 5 Detailed **Methods** and **Materials**
 - 6 Clear & concise **Tables/Figures**
 - 7 **Results** & effective **Discussion**
 - 8 Strong **Conclusion**
 - 9 Correct **References**
 - 10 **Acknowledgments** (if any)



Check out our YouTube channel for effective tips on avoiding common errors in your manuscript!



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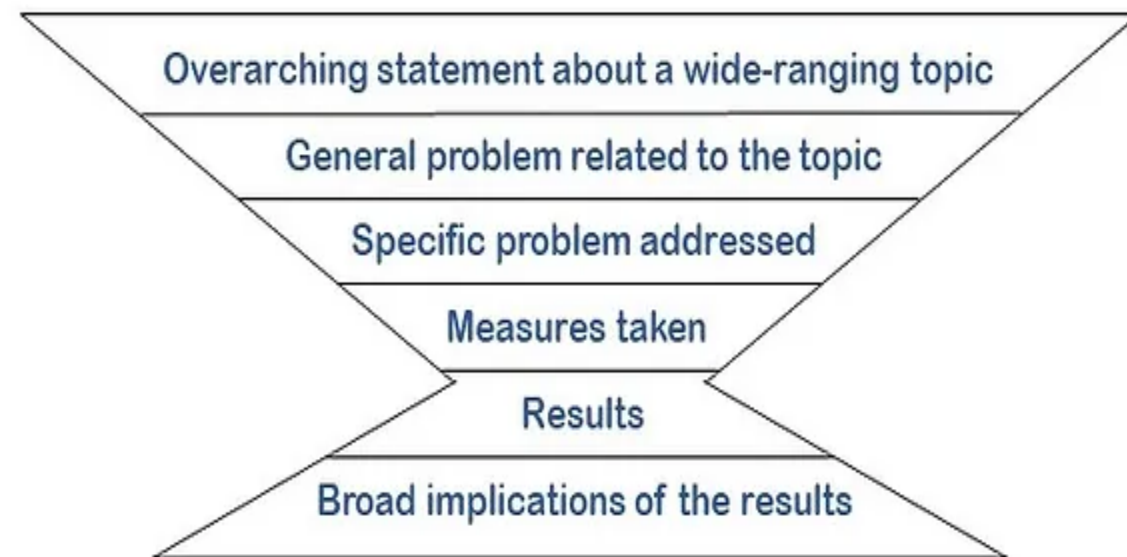


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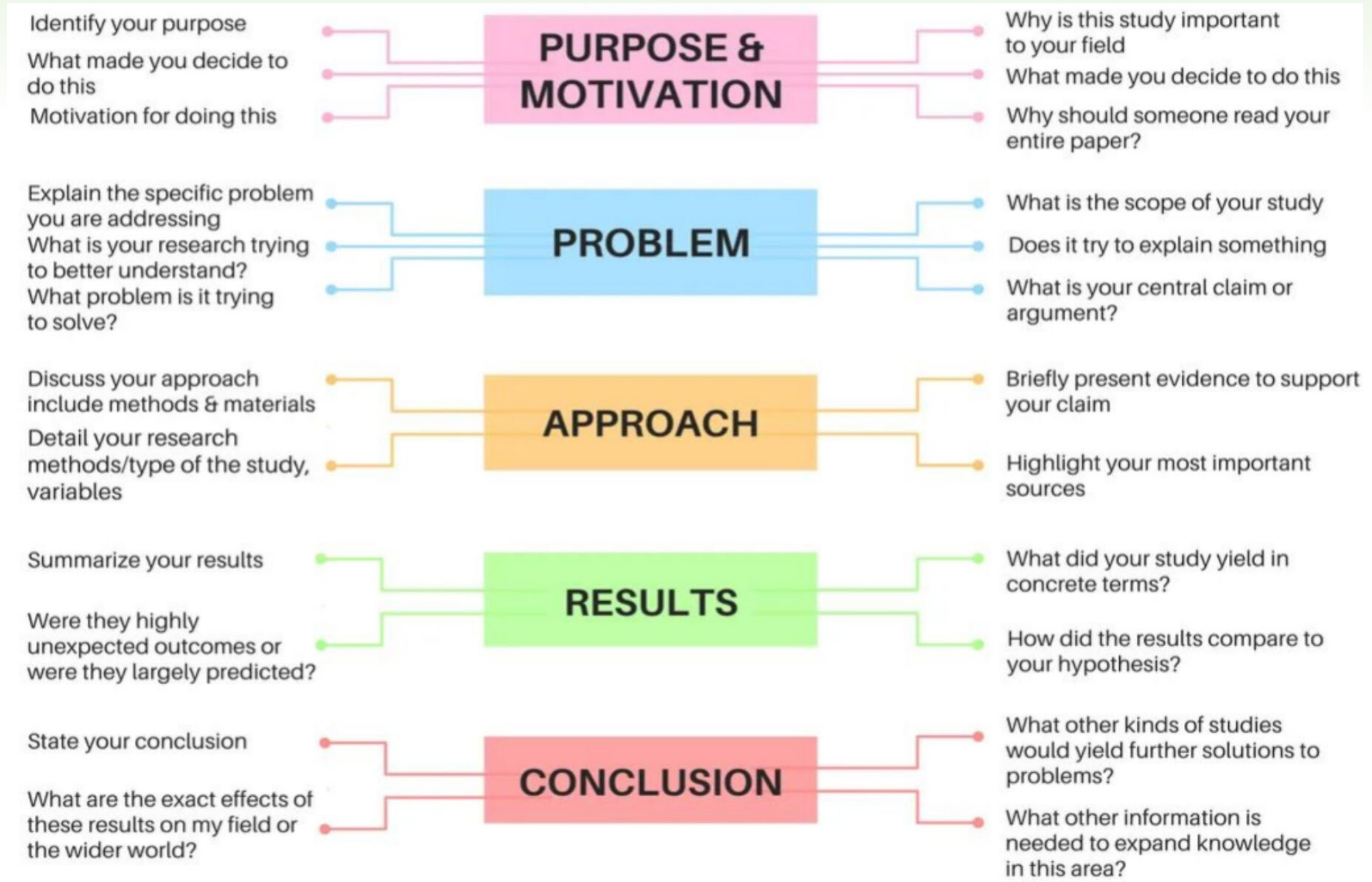
Writing a good abstract

- ❑ Begin with a broad statement about your topic.
- ❑ State the problem or knowledge gap related to this topic that your study explores.
- ❑ Describe what specific aspect of this problem you investigated
- ❑ Briefly explain how you went about doing this.
- ❑ Describe the most meaningful outcome(s) of your study.
- ❑ Finally, close your abstract by explaining the broad implication(s) of your findings.

Abstract Outline

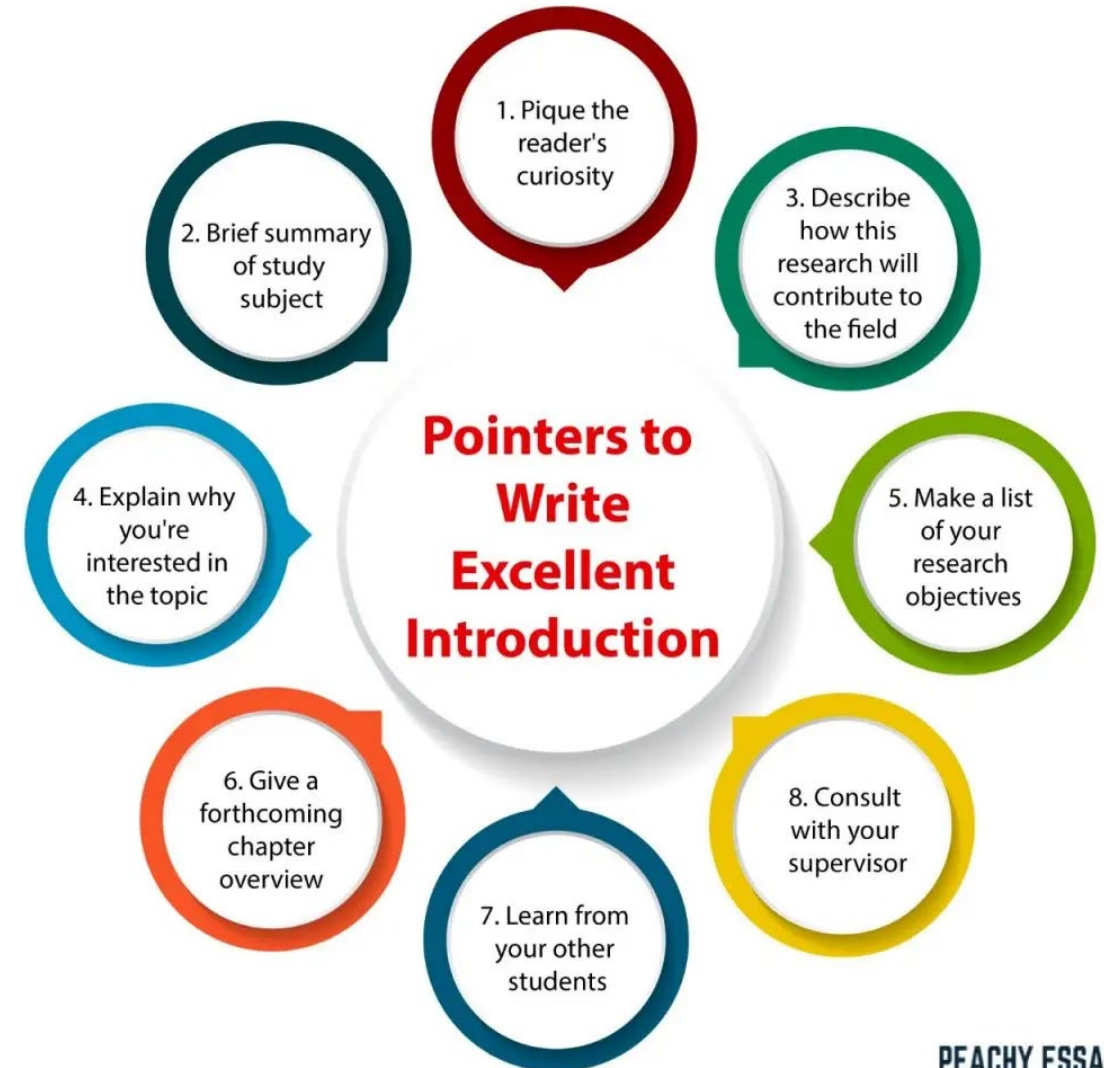


How to write a good abstract



Writing a good introduction

- Draw readers in.
- Zoom in on your topic and its importance.
- Explain how you'll add new knowledge.
- Tell readers what they'll find in your paper.



Methodology section

- Write according to your field of study
- Explain your methodological and data collection approaches in detail
- Ensure that your approaches are suitable for the research problem
- Elaborate on your data collection techniques
- Rationalize your sample size/techniques
- Explain the steps you took to ensure validity and reliability



Writing the result section

Presentation tips

- ✓ Check the instructions or guidelines.
- ✓ Reflect on your research findings.
- ✓ Design your visual presentations.
- ✓ Write your findings section.
- ✓ Edit Your Findings Section.

Practical tips

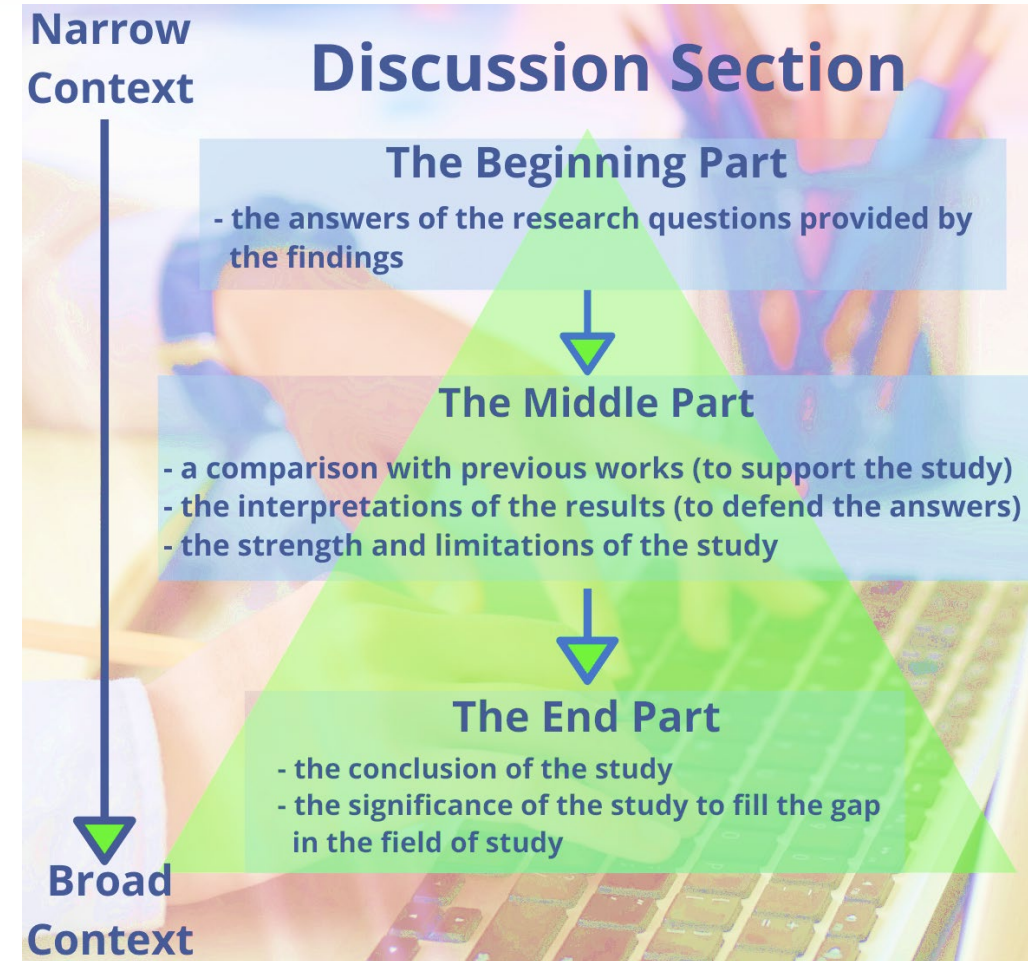
- Summarize your study results
- Present relevant findings only
- Report statistical findings
- Be objective and concise
- Use the correct style to present the findings depending on your study field.



Discussion section

- ❑ **Summary:** A brief recap of your key results
- ❑ **Interpretations:** What do your results mean?
- ❑ **Implications:** Why do your results matter?
- ❑ **Limitations:** What can't your results tell us?
- ❑ **Recommendations:** Avenues for further studies or analyses

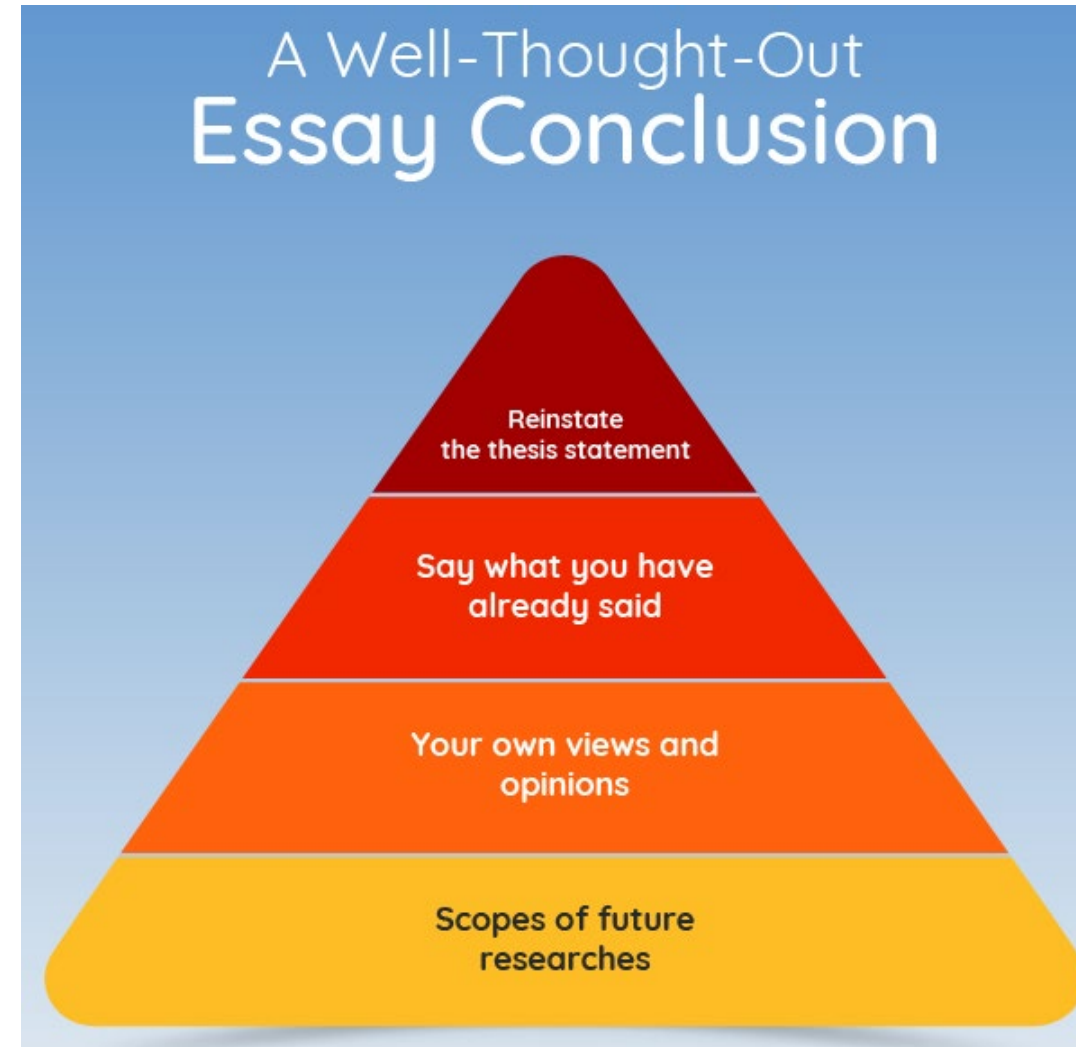
- Don't introduce new results
- Don't make inflated claims
- Don't undermine your research



Writing a Conclusion

A conclusion for a research paper summarizes your key points and the value of your research.

- ❖ Review your thesis statement and research problem
- ❖ Summarize your findings or argument
- ❖ Look toward the future



To be consistent

Discussion Section

Analyzes and interprets research findings in detail

Presents arguments and evidence in a clear and concise manner

Critically evaluates results and explains their significance

Answers questions raised in the introduction and provides recommendations for future research

Includes any limitations or potential flaws in the research

Conclusion Section

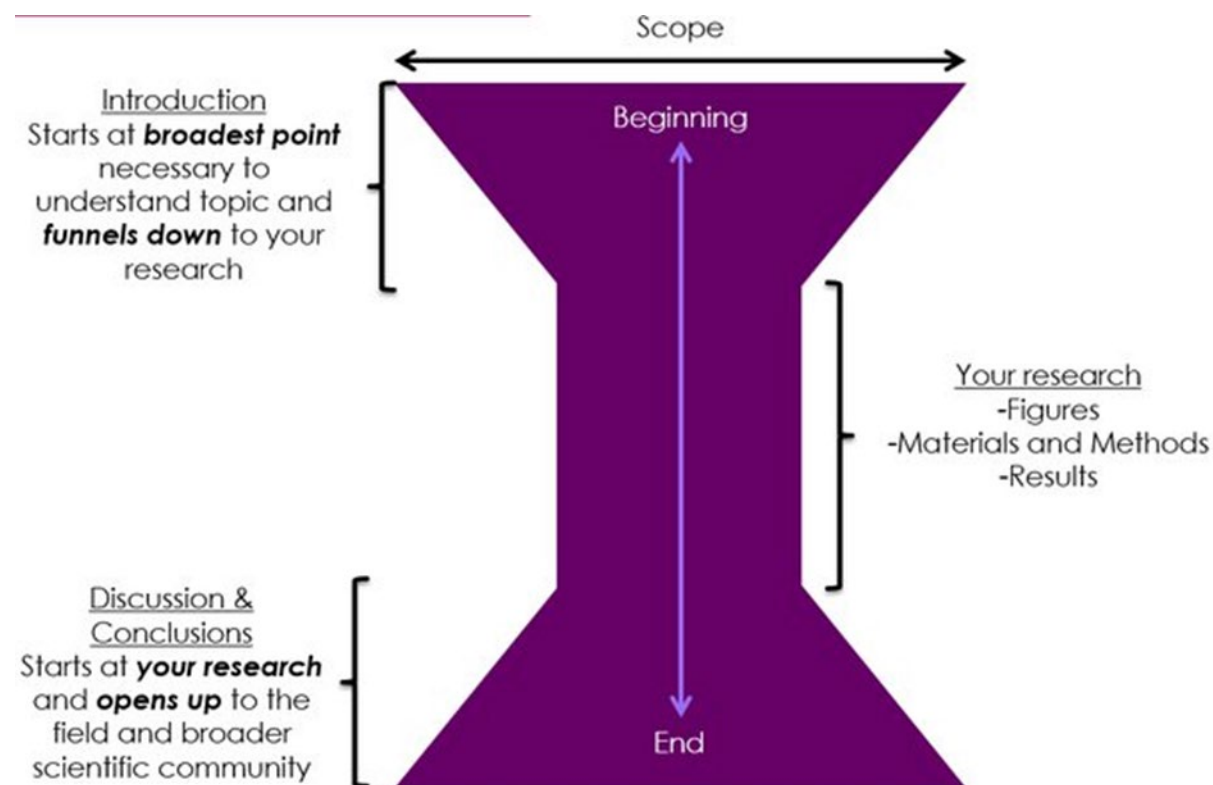
Provides a summary of key points from the thesis

Reinforces the thesis statement

Avoids introducing new information or arguments

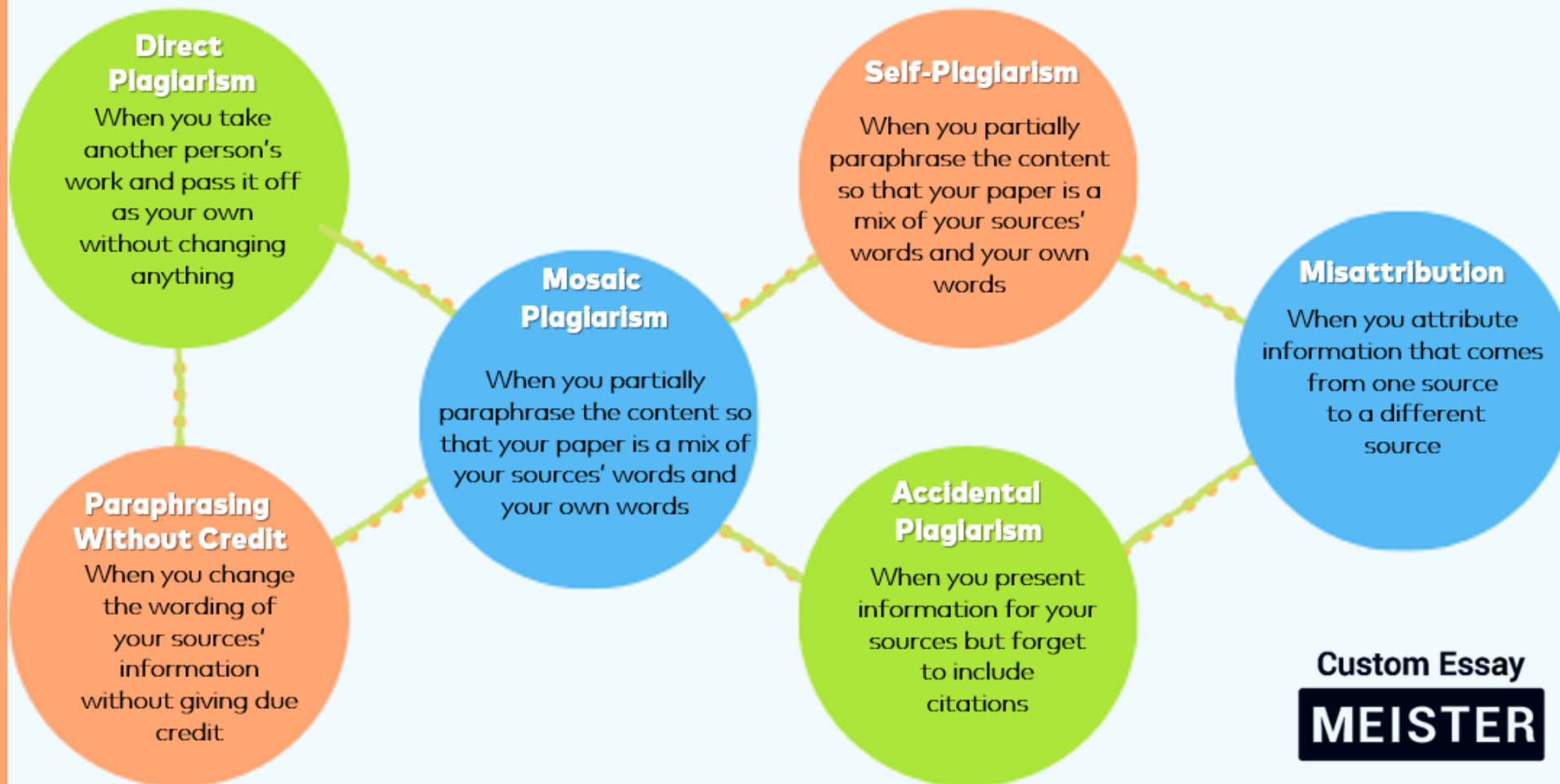
Show relationship between two variables

Summarizes the significance of the research and its contribution to the field



Be safe from the Plagiarism

Types of Plagiarism

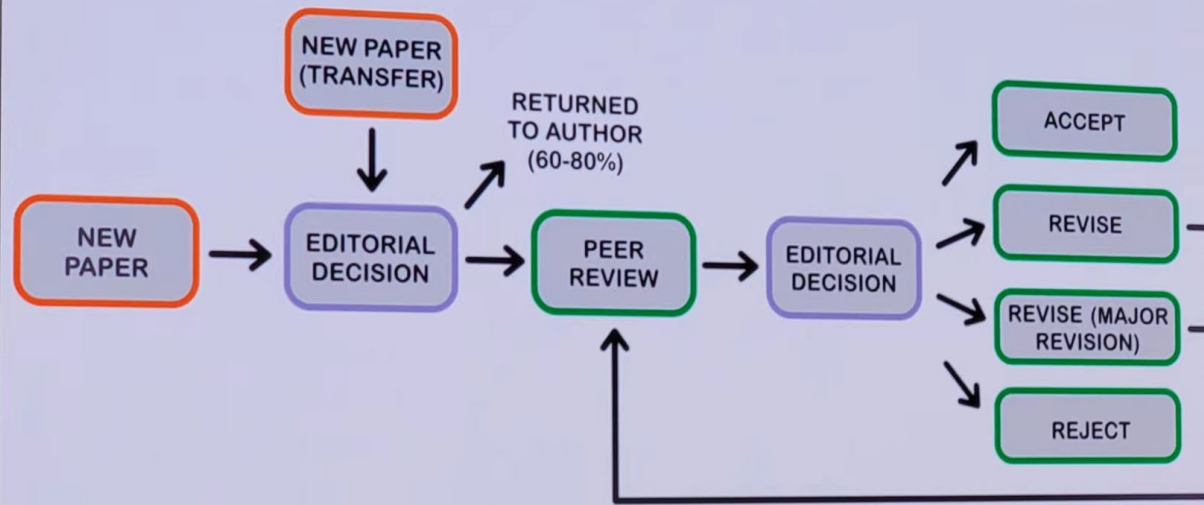


Meeting with the Editor-in-chief of “Nature Reviews: Electrical Engineering” (Personal Experience)

- It is necessary to add some good works that can be considered as the breakthrough in **wind energy harvesting by the TENG research area**
- Writing the manuscript in a different way than the other articles published in non-nature journals.
- Remove the overlapped part with other articles from the manuscript.
- Need more quantitative analysis and benchmark comparisons with more information and data.
- Describing the scaling up of the technology in this field.
- Cost analysis.
- A specific angle or topic of discussion should be maintained.
- Discussion with scientific depth and fundamental breakthrough.
- Commercial application perspective.
- The manuscript should contain information, an educational discussion that can be easily understood by non-technical readers.

Meeting with the Editor of “Nature Nanotechnology” (Personal Experience)

The editorial process



<https://www.nature.com/nature/for-authors/editorial-criteria-and-processes>
<https://www.nature.com/nature/journal-policies/editorial-process>

What are we looking for?

Importance

- **NOVELTY:** new materials/ methods/ insights/ conclusions.
- We compare it to the **published** literature.
(conference abstracts and preprints don't compromise novelty)
- **SIGNIFICANCE:** Of high importance to scientists in the specific field.
- **IMPLICATIONS:** Interesting to researchers in other related disciplines.
- **TECHNICAL ADVANCES** (figure of merits)
- **DEPTH OF KNOWLEDGE**

You better have one paper for point that impress the editor!

Some suggestions

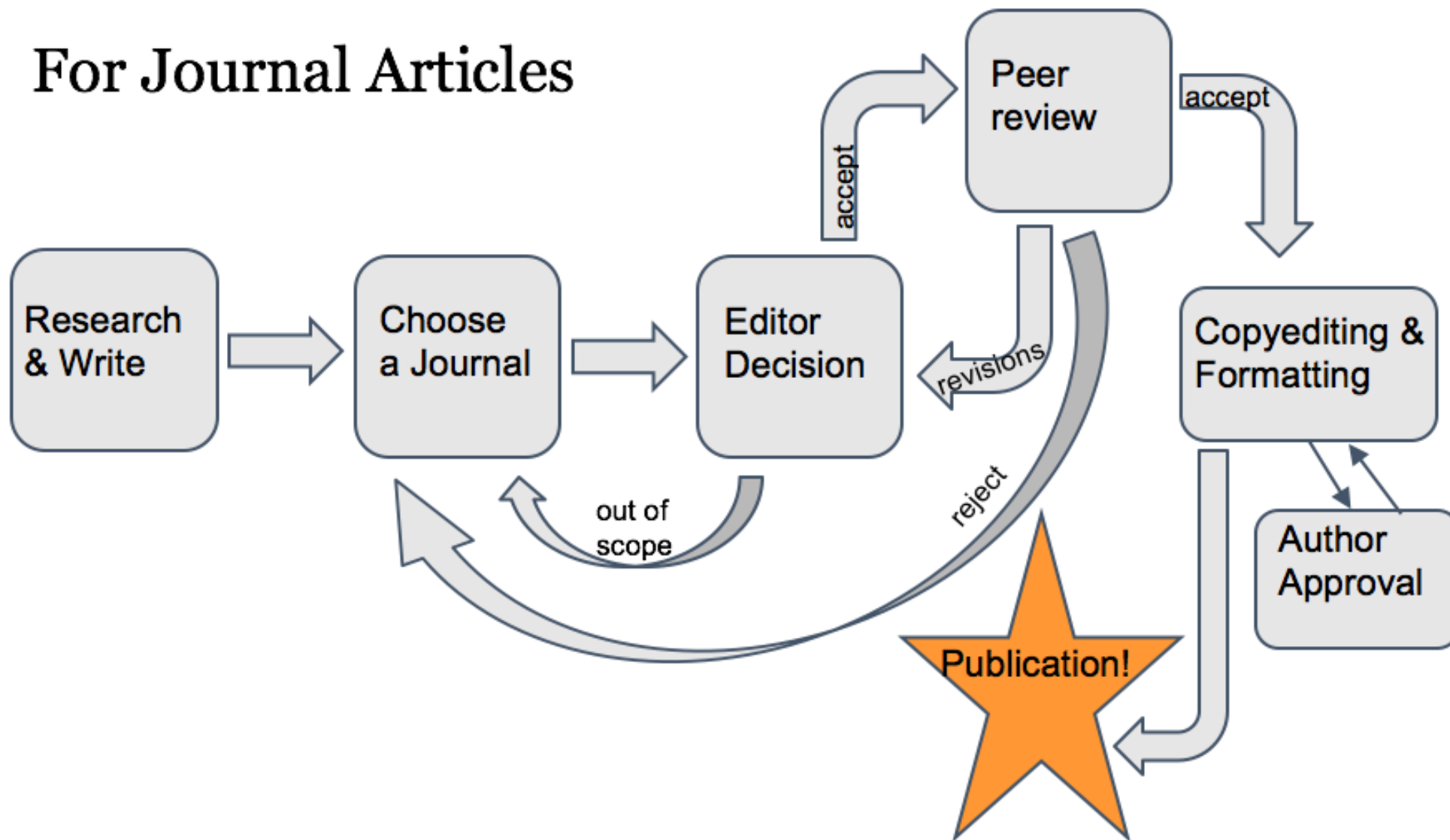
1. 11 steps to structuring a science paper editors will take seriously (<https://www.elsevier.com/connect/11-steps-to-structuring-a-science-paper-editors-will-take-seriously>)
2. Blogs from Grammarly (<https://www.grammarly.com/blog/>)
3. Blogs from Quillbot (<https://quillbot.com/blog/academic-writing-tips/>)

- ❖ Read more journal articles
- ❖ Remember academic reading writing is different from the GRE/IELTS reading writing
- ❖ Try to write as much as possible on blogs, social sites, and everywhere
- ❖ Focus on sentence building and using academic technological terms
- ❖ Just start without waiting, starting is the start of learning

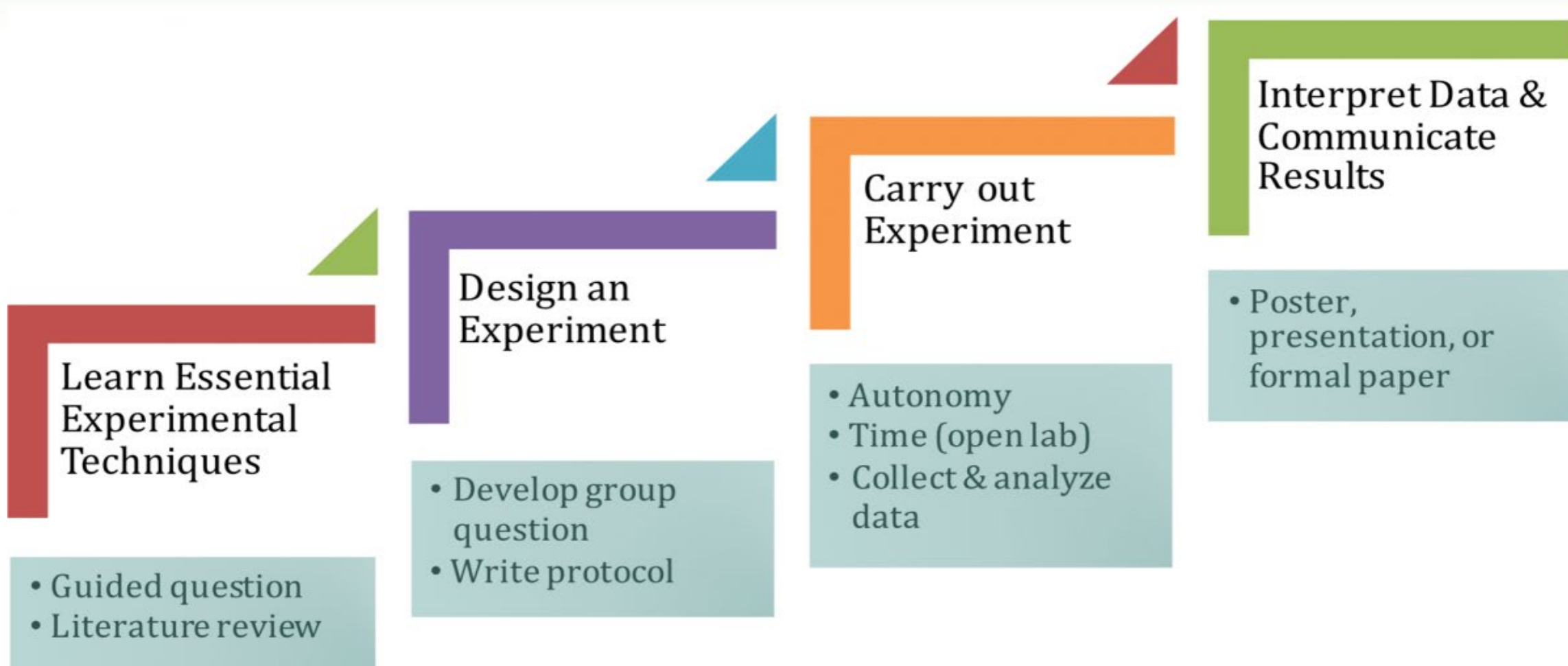


Road to the publication

For Journal Articles

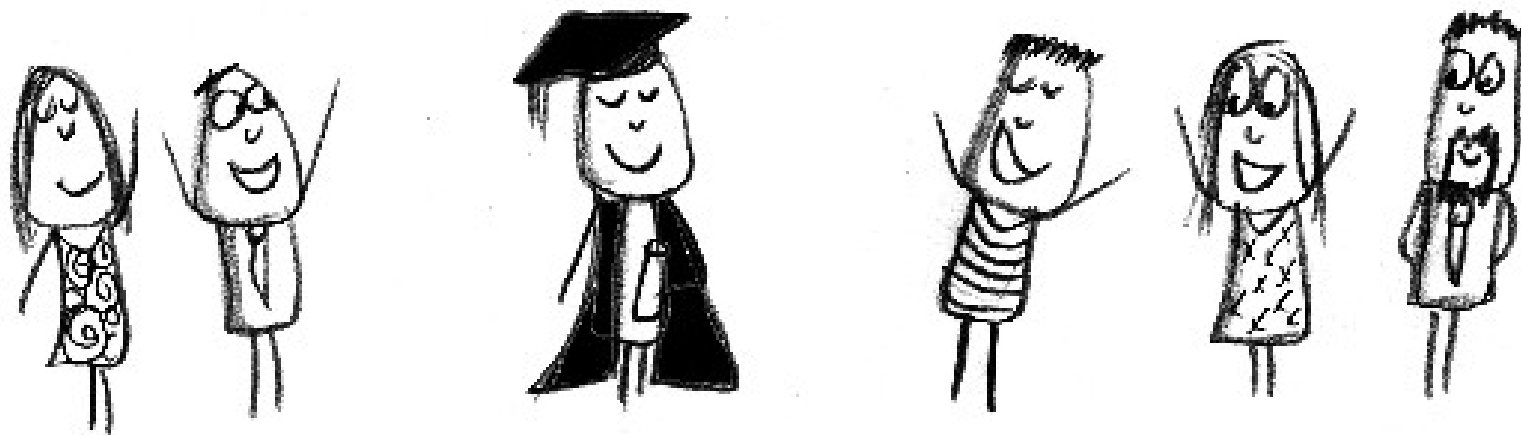


Summarizing the whole process



The instructor/technician(s) act as mentors for steps II-IV

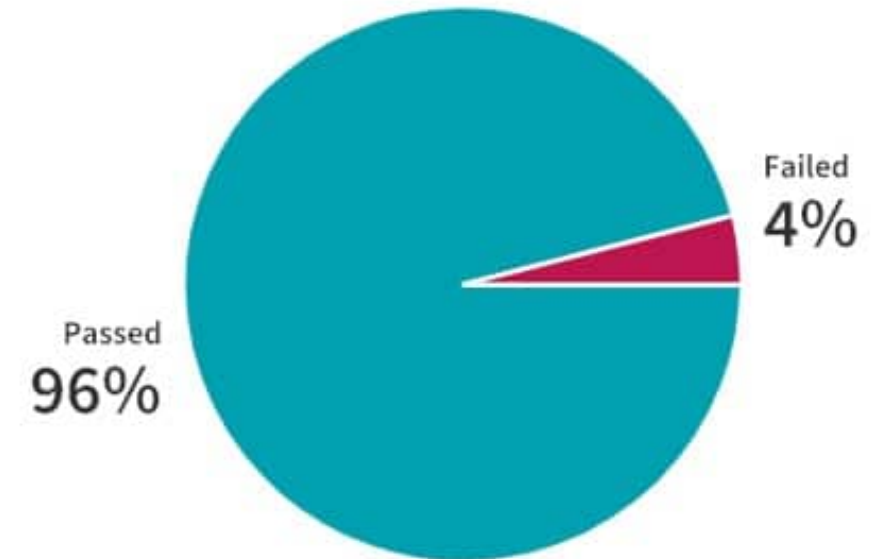
Summary of the talk



No alternative to Research experience



Percentage of students who **passed their viva** (obtained a PhD) vs **failed their viva** (did not obtain a PhD)

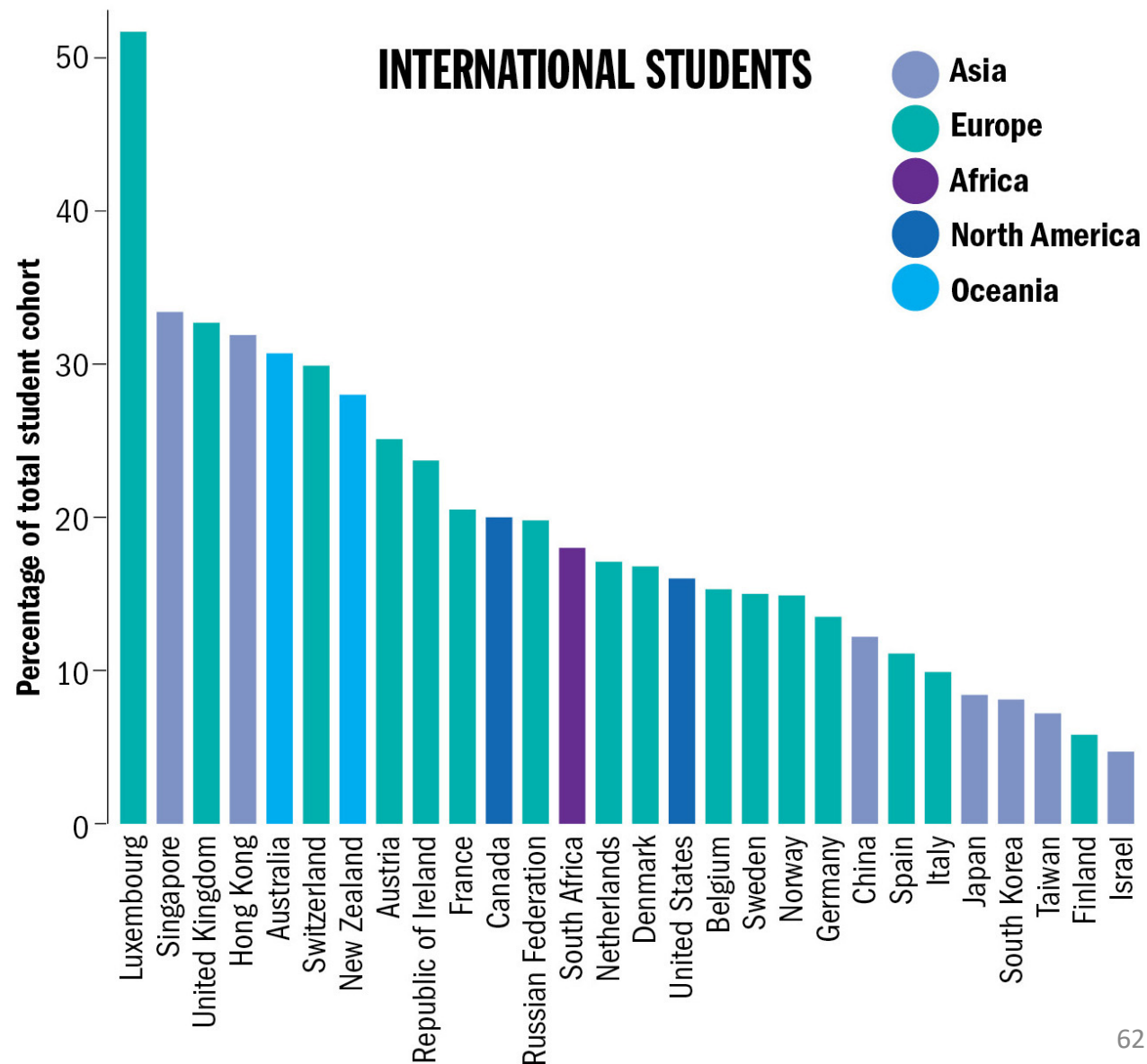
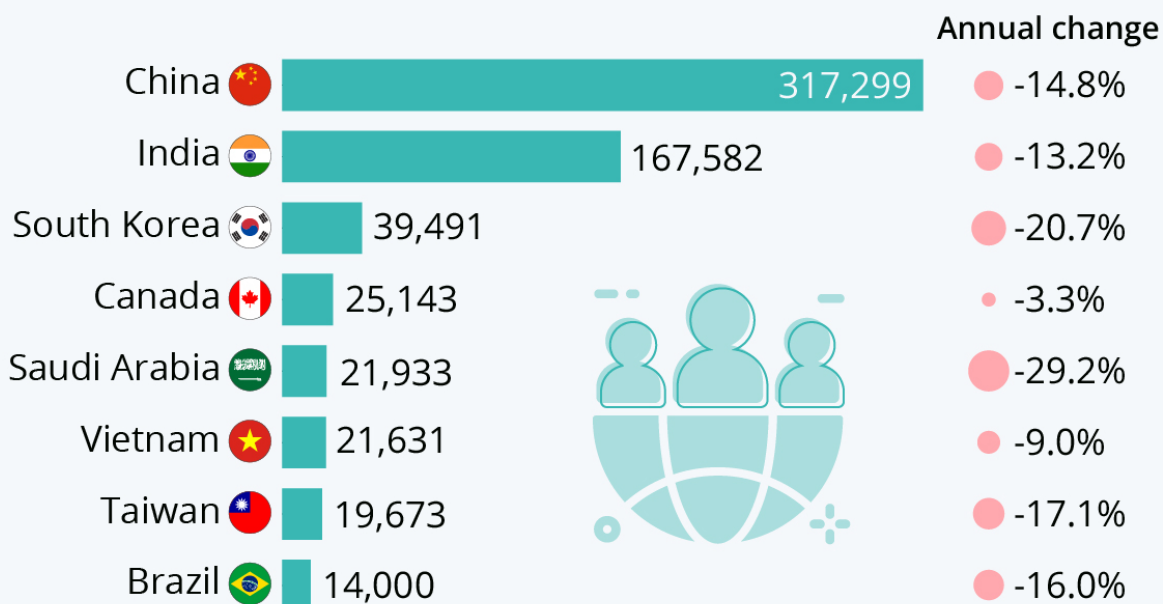


Analysis performed by DiscoverPhDs.com

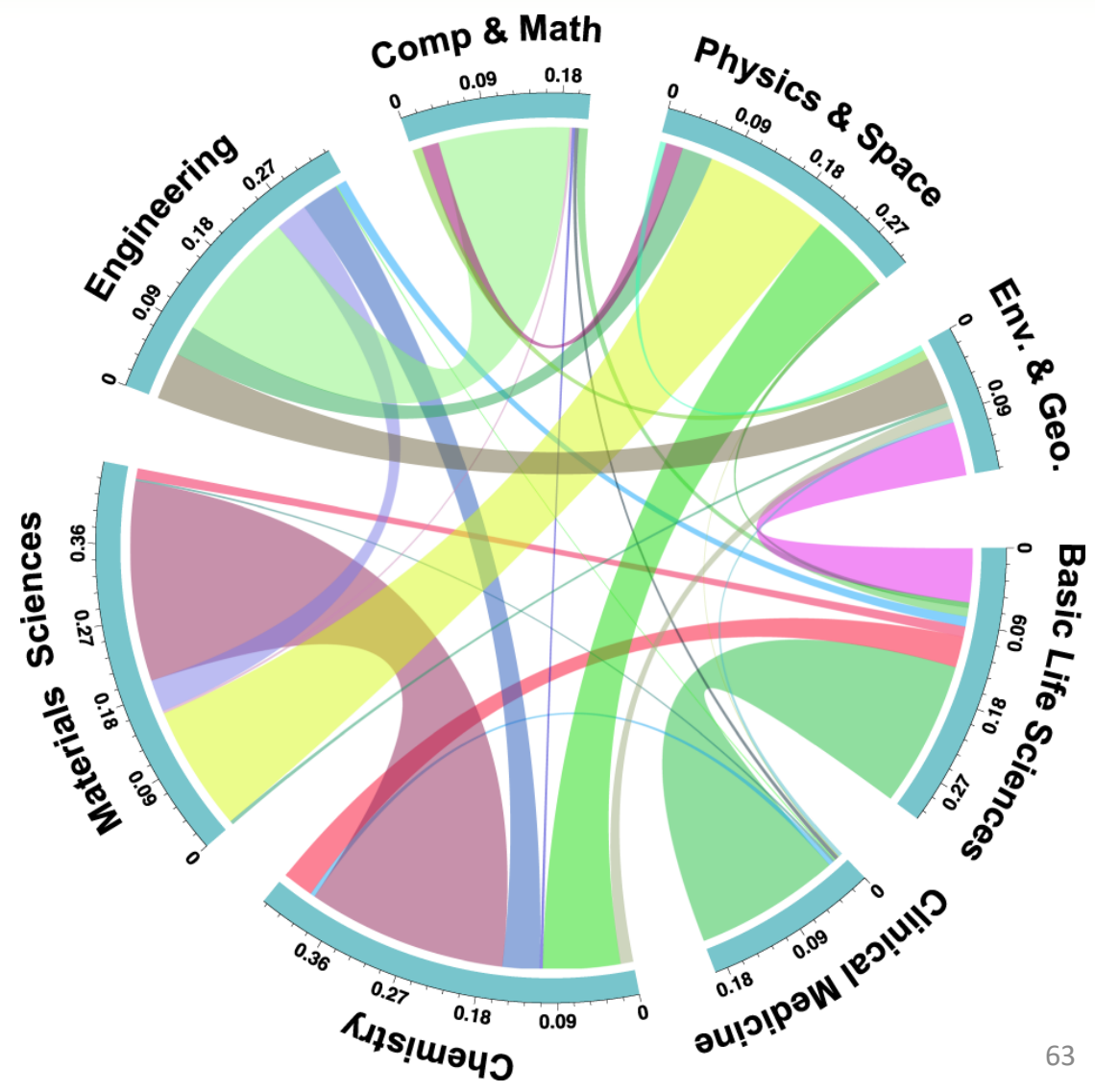
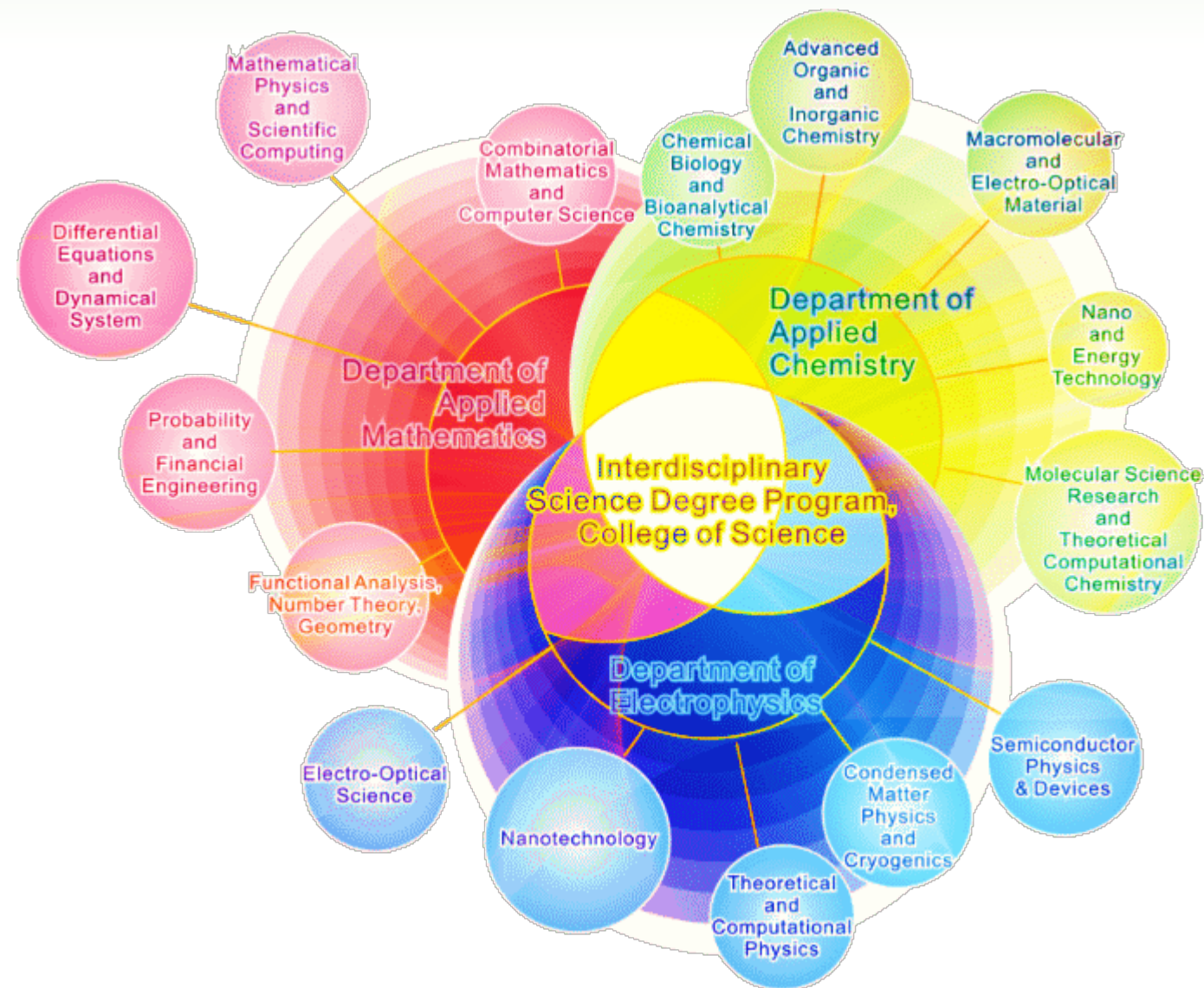
Using multiple routes

Where America's International Students Come From

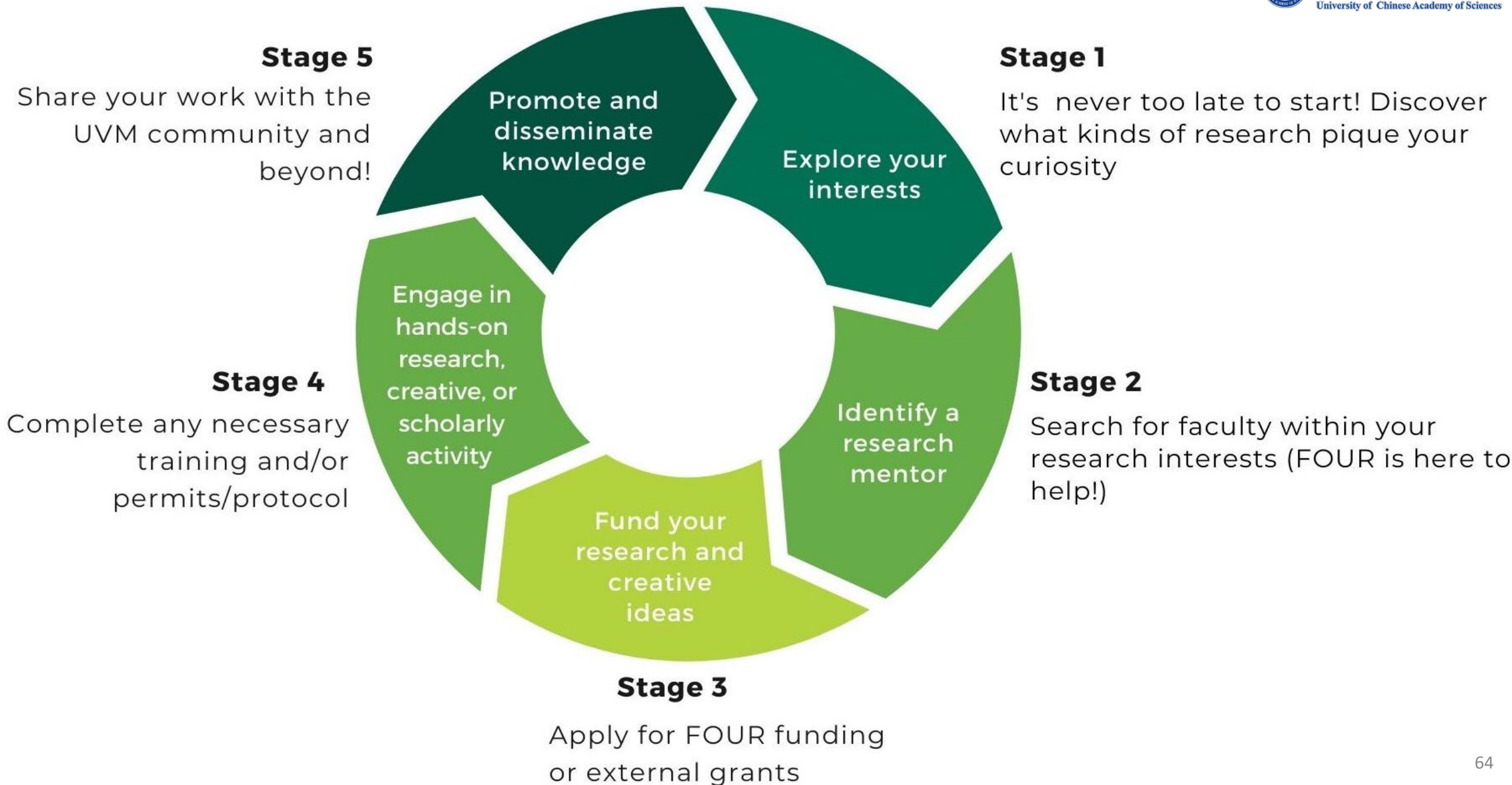
International enrollment in U.S. higher education in the 2020/21 academic year



Using multiple routes



RESEARCH LIFE CYCLE



Road to the success





